A Class of de Bruijn Sequences

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***Abstract*— In this paper, a class of linear feedback shift registers (LFSRs) with characteristic polynomial $(1+x^3)p(x)$ is discussed, where $p(x)$ is a primitive polynomial of degree $n > 2$. The cycle structure and adjacency graphs of the LFSRs are determined. A new class of de Bruijn sequences is constructed from these LFSRs, and the number of de Bruijn sequences in the class is also considered. To illustrate the efficiency of constructing de Bruijn sequences from these LFSRs, an algorithm for producing some corresponding maximum-length nonlinear feedback shift registers with time and memory complexity $O(n)$ is also proposed.**

***Index Terms*— de Bruijn sequence, cycle structure, NFSR, LFSR, cyclotomic number.**

I. Introduction

A de Bruijn sequence of order n is a binary sequence $\mathbf{a} = (a_0, a_1, a_2, \ldots)$ of period 2^n such that all n-tuples $(a_i, a_{i+1}, \ldots, a_{i+n-1})$, $i = 0, 1, \ldots, 2^n - 1$, are pairwise distinct, where n is a positive integer [4]. For example, one period of a certain de Bruijn sequence of order 6 is given by

$$(0000001111011100110110100111010100011001011000101011111100001001). \quad (1)$$

De Bruijn sequences have many applications in communication systems, coding theory and cryptography due to their attractive characteristics such as long periods, large complexities and good randomness properties [6], [14], [15]. Nonlinear feedback shift registers (NFSRs) can be used to generate de Bruijn sequences. Despite of widespread use of NFSRs in modern stream ciphers [5], [18], the theory of NFSRs is far less developed and understood than that of linear feedback shift registers (LFSRs) [8], [14]. One of the challenging problems is to construct maximum-length NFSR sequences, i.e., de Bruijn sequences [4], [12], [14], [30]. For further details of this topic one may refer to an excellent survey by Fredricksen [11].

A known general method of constructing de Bruijn sequences is by joining all the cycles produced by a given feedback shift register (FSR) into a single cycle [11], [14]. Some FSRs with simple cycle structure, such as the maximum-length LFSRs, pure cycling registers and pure summing registers, have been used to generate de Bruijn sequences by the cycle joining method [9]–[11].

Since the cycle structure of an LFSR can be studied systematically, it is natural to construct de Bruijn sequences by applying the cycle joining method to LFSRs. For LFSRs with irreducible characteristic polynomials, Hauge and Helleseth established a connection between the cycles generated by the LFSRs and the irreducible cyclic codes. Consequently, the number of de Bruijn sequences obtained from these LFSRs is related to cyclotomic numbers [16]. For a primitive polynomial $p(x)$ of degree $n \geq 2$, the cycle structure and adjacency graphs of LFSRs with characteristic polynomials of the form $(1+x)^m p(x)$, $m = 1, 2, 3$, have been determined and several classes of de Bruijn sequences are constructed from these LFSRs [20], [24], [29].

Another general way to construct de Bruijn sequences is to use previously known ones as building blocks. Some constructions have been proposed by producing new de Bruijn sequences from those with smaller orders [2], [7], [13], [27], [29], [32]. Most of these constructions are essentially based on the D-morphism developed by Lempel [23], and some of them are generalized to the nonbinary case in [1]. In addition, by determining the cross-join pairs in a de Bruijn sequence of order n, where n is a positive integer, more de Bruijn sequences of order n can be obtained [10], [11]. However, it is difficult to investigate the number and positions of cross-join pairs in an arbitrary de Bruijn sequence. Helleseth and Kløve determined the number of cross-join pairs in an m-sequence of order n (a sequence of period $2^n - 1$ generated by an n-stage LFSR) [19].

The purpose of this paper is to construct new de Bruijn sequences based on LFSRs with reducible characteristic polynomials. To this end, we characterize the cycle structure of a class $\mathcal{F}$ (see Section III) of LFSRs, and determine the adjacency graphs of the LFSRs in $\mathcal{F}$. By means of the cycle joining method, for any integer n with $n > 2$, a new class of de Bruijn sequences of order $n+3$ is constructed, and the size of the proposed class is $O(2^{4n})$ for odd n and $O(2^{8n})$ for even n. Moreover, to illustrate the efficiency of generating de Bruijn sequences from LFSRs in $\mathcal{F}$, we propose an algorithm with time and memory complexity $O(n)$, which can be used to produce $O(2^n)$ $(n+3)$-stage maximum-length NFSRs. For obtaining all the de Bruijn sequences, we sketch an approach to generalizing the algorithm at the end of Section V, and the complexity is also discussed.

In the process of analyzing the adjacency graphs of the LFSRs in $\mathcal{F}$, we characterize the consecutive states of an m-sequence by matrix form. This enables us to transform the state equations for some conjugate pairs into equations over finite fields. With the link between the number of solutions to these equations and the cyclotomic number, some results

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about the cyclotomic number are used for determining the adjacency graphs.

The remainder of this paper is organized as follows. In Section II, we introduce some basic concepts and results. In Section III, the cycle structure of a class of LFSRs is characterized. In Section IV, the adjacency graphs of the LFSRs are given. In Section V, we consider the enumeration and generation of de Bruijn sequences. The final section concludes the paper.

II. Preliminaries

For a positive integer n, let $\mathbb{F}_2^n$ be the n-dimensional vector space over $\mathbb{F}_2$, where $\mathbb{F}_2$ is the finite field with two elements.

A. FSR Sequences

An n-stage *feedback shift register (FSR)* is a circuit consisting of n binary storage elements (called *stages*) regulated by a single clock. The stages are labeled from 0 to $n-1$. A *state* of an FSR is a vector $(x_0, x_1, \ldots, x_{n-1})$, where x_i indicates the content of stage i. The initially loaded content $\mathbf{A}_0 = (a_0, a_1, \ldots, a_{n-1})$ is called an *initial state*. Let f be the feedback function of an FSR. At every clock pulse, for some integer $i \geq 0$, the state $\mathbf{A}_i = (a_i, a_{i+1}, \ldots, a_{i+n-1})$ is updated to $\mathbf{A}_{i+1} = (a_{i+1}, \ldots, a_{i+n-1}, f(a_i, a_{i+1}, \ldots, a_{i+n-1}))$. After consecutive clock pulses, the FSR outputs a sequence $\mathbf{a} = (a_0, a_1, a_2, \ldots)$, called an *output sequence* of the FSR. Therefore, the sequence satisfies the recursion $a_{i+n} = f(a_i, a_{i+1}, \ldots, a_{i+n-1})$ for all $i \geq 0$ and a given initial state $\mathbf{A}_0 = (a_0, a_1, \ldots, a_{n-1})$. We denote by $\Omega(f)$ the set of all 2^n sequences generated by f.

An FSR is called a *linear feedback shift register (LFSR)* if its feedback function f is linear and a *nonlinear feedback shift register (NFSR)* otherwise. The *characteristic polynomial* associated with an n-stage LFSR with feedback function $f(x_0, x_1, \ldots, x_{n-1}) = c_0x_0 + c_1x_1 + \cdots + c_{n-1}x_{n-1}$ is the univariate polynomial $c(x) = x^n + \sum_{i=0}^{n-1} c_i x^i$, where $c_0, \ldots, c_{n-1} \in \mathbb{F}_2$. The output sequences of an LFSR are called *LFSR sequences*, and their properties can be analyzed in terms of the characteristic polynomial. In the sequel, for an LFSR with feedback function f and characteristic polynomial $c(x)$, the set $\Omega(f)$ is also denoted by $\Omega(c(x))$.

The period is an important property of FSR sequences. We first recall some related concepts and results. Let $V(\mathbb{F}_2)$ be the set of all binary infinite sequences. For $\mathbf{a} = (a_0, a_1, a_2, \ldots) \in V(\mathbb{F}_2)$, we define the *shift operator* L on $V(\mathbb{F}_2)$ by

$$L\mathbf{a} = L(a_0, a_1, a_2, \ldots) = (a_1, a_2, a_3, \ldots).$$

Generally, we have $L^m\mathbf{a} = (a_m, a_{m+1}, a_{m+2}, \ldots)$ for any integer m with $m \geq 2$, and denote $L^0\mathbf{a} = \mathbf{a}$ [15]. A sequence $\mathbf{a} = (a_0, a_1, a_2, \ldots) \in V(\mathbb{F}_2)$ is called *periodic* if there exists a positive integer r such that $L^r\mathbf{a} = \mathbf{a}$, i.e., $a_{i+r} = a_i$ for each integer $i \geq 0$. The smallest positive integer r satisfying this property is called the *period* of $\mathbf{a}$, denoted by $\mathrm{per}(\mathbf{a})$. We shall denote $\mathbf{a} = (a_0a_1 \ldots a_{l-1})$ if $\mathbf{a}$ is a periodic sequence with $\mathrm{per}(\mathbf{a}) = l$.

In the remainder of this subsection, suppose that f is the feedback function of an n-stage FSR. It is well known that all sequences in $\Omega(f)$ are periodic if and only if the feedback function f is nonsingular, i.e., f can be written as $f(x_0, x_1, \ldots, x_{n-1}) = x_0 + g(x_1, \ldots, x_{n-1})$ for some function g from $\mathbb{F}_2^{n-1}$ to $\mathbb{F}_2$ [14]. The feedback functions discussed in this paper are always assumed to be nonsingular.

To investigate the period distribution of sequences in $\Omega(f)$, the definition of a cycle in $\Omega(f)$ is recalled from [28]. Two periodic sequences $\mathbf{a}_1$ and $\mathbf{a}_2$ are called *shift-equivalent*, i.e., $\mathbf{a}_1 \simeq \mathbf{a}_2$, if there exists an integer r such that $\mathbf{a}_1 = L^r\mathbf{a}_2$. It is well known that $\simeq$ is an equivalence relation on $\Omega(f)$ [28]. Each shift-equivalent class of $\Omega(f)$ is called a *cycle* in $\Omega(f)$. For a sequence $\mathbf{a} \in \Omega(f)$, denote the cycle in $\Omega(f)$ containing $\mathbf{a}$ by $[\mathbf{a}]$, i.e., $[\mathbf{a}] = \{L^i\mathbf{a} \,|\, i \geq 0\} = \{L^i\mathbf{a} \,|\, 0 \leq i < \mathrm{per}(\mathbf{a})\}$ [28]. Note that the definition of a cycle given here is equivalent to that in [14], which is given in terms of the state graph.

If $\Omega(f)$ consists of exactly r cycles $[\mathbf{a}_1], [\mathbf{a}_2], \ldots, [\mathbf{a}_r]$ where r is a positive integer, then its *cycle structure* is denoted as $\Omega(f) = [\mathbf{a}_1] \cup [\mathbf{a}_2] \cup \cdots \cup [\mathbf{a}_r]$. The corresponding FSR is called a *maximum-length NFSR* if $r = 1$, and its output sequences are the de Bruijn sequences of order n. For LFSRs, we have $r \geq 2$ and the corresponding LFSR is called a *maximum-length LFSR* if $r = 2$. A nonzero output sequence of a maximum-length n-stage LFSR is called an *m-sequence of order n. An LFSR generates m-sequences* if and only if its characteristic polynomial is primitive [14]. Moreover, if the characteristic polynomial $c(x)$ is primitive, then any nonzero sequence $\mathbf{a}$ in $\Omega(c(x))$ has period $2^n - 1$ and $\Omega(c(x)) = [\mathbf{a}] \cup [\mathbf{0}]$, where $\mathbf{0}$ is the all-zero sequence.

Example 1: Let $n = 3$ and the LFSR have primitive characteristic polynomial $x^3 + x^2 + 1 \in \mathbb{F}_2[x]$. Note that (1101001) is an m-sequence in $\Omega(x^3 + x^2 + 1)$. Thus, we have $\Omega(x^3 + x^2 + 1) = [(1101001)] \cup [\mathbf{0}]$.

B. The Cycle Joining Method and Adjacency Graphs

Suppose that f is the feedback function of an n-stage FSR and $\mathbf{a} = (a_0a_1 \ldots a_{l-1}) \in \Omega(f)$, where $\mathbf{a}$ is a periodic sequence with $\mathrm{per}(\mathbf{a}) = l$. Denote $\mathbf{A}_i = (a_i, a_{i+1}, \ldots, a_{i+n-1})$, where i is an integer with $0 \leq i \leq l-1$ and all subscripts are taken modulo l. Note that the $\mathbf{A}_i$'s $(0 \leq i \leq l-1)$ are consecutive states of the FSR when the sequence $\mathbf{a}$ is outputted. For ease of our analysis, it is natural to call $\mathbf{A}_i$ the *i-th state* of $\mathbf{a}$ for $i \geq 1$, and $\mathbf{A}_0$ the *initial state*. Therefore, $\mathbf{a}$ can be represented by an ordered l-tuple, i.e., $\mathbf{a} = (\mathbf{A}_0\mathbf{A}_1 \ldots \mathbf{A}_{l-1})$. Among the states, $\mathbf{A}_{i+1}$ is the *successor* of $\mathbf{A}_i$ for $0 \leq i \leq l-2$ and $\mathbf{A}_0$ is the *successor* of $\mathbf{A}_{l-1}$. Note that $[\mathbf{a}] = [L\mathbf{a}] = \cdots = [L^{l-1}\mathbf{a}]$. Then $[(\mathbf{A}_0\mathbf{A}_1 \ldots \mathbf{A}_{l-1})]$, $[(\mathbf{A}_1\mathbf{A}_2 \ldots \mathbf{A}_{l-1}\mathbf{A}_0)]$, …, and $[(\mathbf{A}_{l-1}\mathbf{A}_0 \ldots \mathbf{A}_{l-2})]$ denote the same cycle. In the sequel, for each $0 \leq i \leq l-1$ we also call $\mathbf{A}_i$ *a state on* the cycle $[\mathbf{a}]$.

Let $\mathbf{X}$ be a state of the cycle $[\mathbf{a}]$ in $\Omega(f)$. We define the *conjugate* $\widehat{\mathbf{X}}$ and the *complement* $\overline{\mathbf{X}}$ of the n-tuple $\mathbf{X} = (x_0, x_1, \ldots, x_{n-1})$ by

$$\widehat{\mathbf{X}} = (\overline{x}_0, x_1, \ldots, x_{n-1}), \quad \overline{\mathbf{X}} = (\overline{x}_0, \overline{x}_1, \ldots, \overline{x}_{n-1}), \qquad (2)$$

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Fig. 1. The adjacency graph of the LFSR with characteristic polynomial $x^3 + x^2 + 1$.

where $\overline{x}_i$ is the binary complement of x_i. The two states $\mathbf{X}$ and $\widehat{\mathbf{X}}$ form *a conjugate pair*. Two cycles [**a**] and [**b**] are said to be *adjacent* if they are disjoint and there exists a state $\mathbf{X}$ on [**a**] whose conjugate $\widehat{\mathbf{X}}$ is on [**b**]. It is well-known that two adjacent cycles C_1 and C_2 are joined into a single cycle when the successors of $\mathbf{X}$ and $\widehat{\mathbf{X}}$ are interchanged. This is the basic idea of the *cycle joining method* introduced in [14]. The problem of determining conjugate pairs between cycles in a state graph leads us to the definition of the adjacency graph.

Definition 1 ([17], [26]): For an FSR with feedback function f, its *adjacency graph* is an undirected multigraph where the vertices correspond to the cycles in $\Omega(f)$, and there exists an edge between two vertices if and only if they share a conjugate pair.

Two vertices can be connected by more than one edge in an adjacency graph. In this case, the edge between two vertices sharing exactly m conjugate pairs can be labeled with an integer m [11]. For instance, the LFSR in Example 1 generates exactly two cycles [(1101001)] and [**0**]. It is easy to verify that the two cycles [(1101001)] and [**0**] share exactly one conjugate pair $\{(1, 0, 0), (0, 0, 0)\}$. Then, by Definition 1, the adjacency graph consists of two vertices and one edge labeled with 1, as shown in Fig. 1.

To characterize the de Bruijn sequences constructed by the cycle joining method, the concept of spanning trees of a graph is of significance. Let G be an undirected graph. A connected undirected subgraph T of G is called a *spanning tree* of G if T contains no cycles (in the sense of graph theory) and includes every vertex of G [3]. Now we can give the following fact.

Fact 1 ([16], [17]): There is a one-to-one correspondence between the spanning trees of an adjacency graph and the de Bruijn sequences constructed by the cycle joining method.

C. Cyclotomy

For a positive integer n, let α be a primitive element of $\mathbb{F}_{2^n}$, where $\mathbb{F}_{2^n}$ is the finite field with 2^n elements. Denote

$$J = \{0, 1, \ldots, 2^n - 2\} \quad \text{and} \quad J^* = J \backslash \{0\}. \tag{3}$$

For any $j \in J^*$, let Z be a mapping from J^* to J^* such that

$$\alpha^{Z(j)} = 1 + \alpha^j. \tag{4}$$

Note that the mapping Z is a permutation of J^* [15].

Let $d \geq 1$ be a divisor of $2^n - 1$. The *cyclotomic classes* $C_0, C_1, \ldots, C_{d-1}$ of $\mathbb{F}_{2^n}$ are defined by

$$C_i = \left\{ \alpha^{i+jd} \,\middle|\, 0 \leq j \leq \frac{2^n - 1}{d} - 1 \right\},$$

where $0 \leq i \leq d-1$. For integers l, m with $0 \leq l, m \leq d-1$, the *cyclotomic number* $(l, m)_d$ over $\mathbb{F}_{2^n}$ is defined as the number of elements $x \in C_l$ such that $1 + x \in C_m$. A connection between cyclotomic numbers and the permutation Z is given in the following lemma.

Lemma 1: With the notation defined in this subsection, $(l, m)_d = \left|\left\{(j, Z(j)) \,\middle|\, j \equiv l \,(\mathrm{mod}\, d),\ Z(j) \equiv m \,(\mathrm{mod}\, d),\ j \in J^*\right\}\right|$, where $l, m \in \{0, 1, \ldots, d-1\}$, and $(l, m)_d$ is a cyclotomic number of order d.

Proof: By the definition of a cyclotomic number, $(l, m)_d = \left|\left\{x \in \mathbb{F}_{2^n} \,\middle|\, x \in C_l,\ 1 + x \in C_m\right\}\right|$. For each pair (l, m) with $l, m \in \{0, 1, \ldots, d-1\}$, since neither 0 nor 1 in $\mathbb{F}_{2^n}$ is counted in $(l, m)_d$, it follows that $(l, m)_d = \left|\left\{x \in \mathbb{F}_{2^n} \backslash \{0, 1\} \,\middle|\, x \in C_l,\ 1 + x \in C_m\right\}\right|$. Then we have

$$\begin{aligned}
&(l, m)_d \\
&= \left|\left\{x \in \mathbb{F}_{2^n} \backslash \{0, 1\} \,\middle|\, x \in C_l,\ 1 + x \in C_m\right\}\right| \\
&= \left|\left\{j \in J^* \,\middle|\, \alpha^j \in C_l,\ 1 + \alpha^j \in C_m\right\}\right| \\
&= \left|\left\{j \in J^* \,\middle|\, \alpha^j \in C_l,\ \alpha^{Z(j)} \in C_m\right\}\right| \\
&= \left|\left\{(j, Z(j)) \,\middle|\, j \equiv l \,(\mathrm{mod}\, d),\ Z(j) \equiv m \,(\mathrm{mod}\, d),\ j \in J^*\right\}\right|
\end{aligned}$$

as claimed. ∎

Note that $3 \mid (2^n - 1)$ for an even n. Thus, for even n, the cyclotomic numbers of order 3 over $\mathbb{F}_{2^n}$ can be defined. The explicit values of these cyclotomic numbers are given in the following lemma, which will be used in the sequel.

Lemma 2 ([16], [33]): The cyclotomic numbers of order 3 are given by

$$A = \frac{2^n + (-2)^{\frac{n}{2}+1} - 8}{9}, \quad B = C = \frac{2^n + (-2)^{\frac{n}{2}} - 2}{9}$$

and

$$D = \frac{2^n + (-2)^{\frac{n}{2}+1} + 1}{9},$$

where $A = (0, 0)_3$, $B = (0, 1)_3 = (1, 0)_3 = (2, 2)_3$, $C = (0, 2)_3 = (2, 0)_3 = (1, 1)_3$ and $D = (1, 2)_3 = (2, 1)_3$.

III. The Cycle Structure of a Class $\mathcal{F}$ of LFSRs

Let $\mathcal{F}$ denote the set of LFSRs with characteristic polynomials of the form

$$q(x) = (1 + x^3)p(x), \tag{5}$$

where $p(x)$ is a primitive polynomial of degree n over $\mathbb{F}_2$ and n is an integer with $n > 2$. The cycle structure of each LFSR in $\mathcal{F}$ will be determined in this section.

We will recall some classic results on the cycle structure of LFSRs from [31] and [35]. Let $\mathbf{a} = (a_0, a_1, a_2, \ldots)$ and $\mathbf{b} = (b_0, b_1, b_2, \ldots)$ be two sequences in $V(\mathbb{F}_2)$ and $c \in \mathbb{F}_2$. The sum $\mathbf{a} + \mathbf{b}$ and scalar multiplication $c\mathbf{a}$ are defined by

$$\begin{aligned}
\mathbf{a} + \mathbf{b} &= (a_0 + b_0, a_1 + b_1, a_2 + b_2, \ldots), \\
c\mathbf{a} &= (ca_0, ca_1, ca_2, \ldots),
\end{aligned} \tag{6}$$

where both the sum $a_i + b_i$ and the multiplication ca_i are operations on $\mathbb{F}_2$ for every $i \geq 0$. Let $f(x)$ be the characteristic polynomial of an n-stage LFSR. Then the set $\Omega(f(x))$ is a vector space over $\mathbb{F}_2$ endowed with the two operations defined in (6) [31]. Some properties of $\Omega(f(x))$ are listed in the following lemma.

Lemma 3 ([35]): For two nonzero monic polynomials $f(x), g(x) \in \mathbb{F}_2[x]$, the following statements hold.

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(i) $\Omega(g(x)) \subset \Omega(f(x))$ if and only if $g(x)|f(x)$.

(ii) Let $\Omega(f(x)) + \Omega(g(x))$ denote the set of sequences $\mathbf{a} + \mathbf{b}$ with $\mathbf{a} \in \Omega(f(x))$ and $\mathbf{b} \in \Omega(g(x))$. Then $\Omega(f(x)) + \Omega(g(x)) = \Omega(\mathrm{lcm}(f(x), g(x)))$. In particular, if $\gcd(f(x), g(x)) = 1$, then $\Omega(f(x)) + \Omega(g(x)) = \Omega(f(x)g(x))$ and $\mathrm{per}(\mathbf{a} + \mathbf{b}) = \mathrm{lcm}(\mathrm{per}(\mathbf{a}), \mathrm{per}(\mathbf{b}))$.

For m-sequences, the so-called shift-and-add property given in the following lemma will be frequently used in the sequel.

Lemma 4 ([15], [35]): If $\mathbf{a}$ is a binary m-sequence of period $2^n - 1$, then for each pair (i, j) with $0 \le i, j < 2^n - 1$ and $i \ne j$, there exists some integer k with $0 \le k < 2^n - 1$ such that

$$L^i\mathbf{a} + L^j\mathbf{a} = L^k\mathbf{a}.$$

By applying Lemmas 3 (ii) and 4, the cycle structure of a given n-stage LFSR in $\mathcal{F}$ is determined in the following proposition.

Proposition 1: Let n be a positive integer with $n > 2$. For a given $(n + 3)$-stage LFSR in $\mathcal{F}$, its cycle structure is given as follows.

(i) If n is odd, we have

$$\begin{aligned}\Omega(q(x)) = {} & [\mathbf{0}] \cup [\mathbf{1}] \cup [\mathfrak{u}] \cup [\overline{\mathfrak{u}}] \cup [\mathfrak{s}] \cup [\overline{\mathfrak{s}}] \\ & \cup [\mathfrak{u} + \mathfrak{s}] \cup [\overline{\mathfrak{u}} + \mathfrak{s}],\end{aligned} \tag{7}$$

where $\mathfrak{u} \in [(100)]$, $\mathfrak{s}$ is an m-sequence in $\Omega(p(x))$, $\mathbf{1}$ is the all-one sequence, $\overline{\mathfrak{u}} = \mathfrak{u} + \mathbf{1}$ and $\overline{\mathfrak{s}} = \mathfrak{s} + \mathbf{1}$.

(ii) If n is even, we have

$$\begin{aligned}\Omega(q(x)) = {} & [\mathbf{0}] \cup [\mathbf{1}] \cup [\mathfrak{u}] \cup [\overline{\mathfrak{u}}] \cup [\mathfrak{s}] \cup [\overline{\mathfrak{s}}] \\ & \bigcup_{i=0}^{2}[L^i\mathfrak{u} + \mathfrak{s}] \bigcup_{i=0}^{2}[L^i\overline{\mathfrak{u}} + \mathfrak{s}],\end{aligned} \tag{8}$$

where $\mathfrak{u} \in [(100)]$, $\mathfrak{s}$ is an m-sequence in $\Omega(p(x))$, $\mathbf{1}$ is the all-one sequence, $\overline{\mathfrak{u}} = \mathfrak{u} + \mathbf{1}$ and $\overline{\mathfrak{s}} = \mathfrak{s} + \mathbf{1}$.

Proof: Let the characteristic polynomial of a given LFSR in $\mathcal{F}$ be $q(x) = (1 + x^3)p(x)$, where $p(x)$ is a primitive polynomial of degree n over $\mathbb{F}_2$ and $n > 2$. Note that $1 + x^3 = (1 + x)(1 + x + x^2)$ and $1 + x + x^2$ is primitive over $\mathbb{F}_2$. Since $p(x)$ is primitive and $p(x) \nmid (1 + x^3)$, we have $\gcd(1 + x^3, p(x)) = 1$. Thus, by Lemma 3 (ii), we have

$$\Omega(q(x)) = \Omega(1 + x^3) + \Omega(p(x)). \tag{9}$$

It is easily seen that $\Omega(1 + x^3) = [\mathbf{0}] \cup [\mathbf{1}] \cup [\mathfrak{u}] \cup [\overline{\mathfrak{u}}]$, where $\mathfrak{u} \in [(100)]$, and $\overline{\mathfrak{u}} = \mathfrak{u} + \mathbf{1}$. On the other hand, it follows from the property of an m-sequence that $\Omega(p(x)) = [\mathbf{0}] \cup [\mathfrak{s}]$, where $\mathfrak{s}$ is an m-sequence in $\Omega(p(x))$. Hence each sequence in $\Omega(q(x))$ can be written as

$$L^i\mathbf{a} + L^j\mathbf{b}, \quad 0 \le i < \mathrm{per}(\mathbf{a}),\ 0 \le j < \mathrm{per}(\mathbf{b}),$$

where $\mathbf{a} \in \{\mathbf{0}, \mathbf{1}, \mathfrak{u}, \overline{\mathfrak{u}}\}$ and $\mathbf{b} \in \{\mathbf{0}, \mathfrak{s}\}$. To determine all cycles in $\Omega(q(x))$, the following two cases are considered.

Case 1: either $\mathrm{per}(\mathbf{a}) = 1$ or $\mathrm{per}(\mathbf{b}) = 1$. If $\mathbf{a} = \mathbf{0}$ or $\mathbf{b}$=$\mathbf{0}$, by Lemma 3 (i), it is clear that the cycles $[\mathbf{0}]$, $[\mathbf{1}]$, $[\mathfrak{u}]$, $[\overline{\mathfrak{u}}]$ and $[\mathfrak{s}]$ are in $\Omega(q(x))$. If $\mathbf{a} = \mathbf{1}$, then $[\overline{\mathfrak{s}}] = [\mathbf{1} + \mathfrak{s}]$ is in $\Omega(q(x))$.

Case 2: $\mathrm{per}(\mathbf{a}) = 3$ and $\mathrm{per}(\mathbf{b}) = 2^n - 1$, i.e., $\mathbf{a} \in \{\mathfrak{u}, \overline{\mathfrak{u}}\}$ and $\mathbf{b} = \mathfrak{s}$.

In the case of $\mathbf{a} = \mathfrak{u}$ and $\mathbf{b} = \mathfrak{s}$, note that

$$\begin{aligned}[L^i\mathfrak{u} + L^j\mathfrak{s}] &= [L^j(L^{(i-j)(\mathrm{mod}\,3)}\mathfrak{u} + \mathfrak{s})] \\ &= [L^{(i-j)(\mathrm{mod}\,3)}\mathfrak{u} + \mathfrak{s}].\end{aligned} \tag{10}$$

When i and j run through the sets $\{0, 1, 2\}$ and $\{0, 1, \ldots, 2^n - 2\}$ respectively, by Equality (10), it is sufficient to consider the cycles $[L^i\mathfrak{u} + \mathfrak{s}]$ for $i \in \{0, 1, 2\}$.

If n is odd, then $2^n \equiv 2 \,(\mathrm{mod}\, 3)$. By Lemma 3 (ii), $\mathrm{per}(L^k\mathfrak{u} + \mathfrak{s}) = \mathrm{lcm}(3, 2^n - 1) = 3(2^n - 1)$. Then we have $[\mathfrak{u}+\mathfrak{s}]=[L^{2^n-1}(\mathfrak{u}+\mathfrak{s})] = [L^{(2^n-1)\,(\mathrm{mod}\,3)}\mathfrak{u}+L^{2^n-1}\mathfrak{s}] = [L\mathfrak{u}+\mathfrak{s}]$ and $[L\mathfrak{u} + \mathfrak{s}] = [L^{2^n-1}(L\mathfrak{u} + \mathfrak{s})] = [L^{2^n\,(\mathrm{mod}\,3)}\mathfrak{u} + L^{2^n-1}\mathfrak{s}] = [L^2\mathfrak{u} + \mathfrak{s}]$. That is, $[\mathfrak{u} + \mathfrak{s}] = [L\mathfrak{u} + \mathfrak{s}] = [L^2\mathfrak{u} + \mathfrak{s}]$. Therefore, in this case, only one cycle $[\mathfrak{u} + \mathfrak{s}]$ in $\Omega(q(x))$ is obtained when i, j run through the sets $\{0, 1, 2\}$ and $\{0, 1, \ldots, 2^n - 2\}$ respectively.

If n is even, then $3 \mid (2^n - 1)$. It follows from Lemma 3 (ii) that $\mathrm{per}(L^k\mathfrak{u}+\mathfrak{s}) = \mathrm{lcm}(3, 2^n-1) = 2^n-1$. We claim that the cycles $[\mathfrak{u}+\mathfrak{s}]$, $[L\mathfrak{u}+\mathfrak{s}]$, $[L^2\mathfrak{u}+\mathfrak{s}]$ are pairwise distinct. That is, for different integers $i_1, i_2 \in \{0, 1, 2\}$, $L^{i_1}\mathfrak{u} + \mathfrak{s} \notin [L^{i_2}\mathfrak{u} + \mathfrak{s}]$. Otherwise, there exists an integer k, $0 \le k < 2^n - 2$, such that

$$L^k(L^{i_1}\mathfrak{u} + \mathfrak{s}) = L^{i_2}\mathfrak{u} + \mathfrak{s}. \tag{11}$$

It is straightforward that $k > 0$ since $L^{i_1}\mathfrak{u} + \mathfrak{s} \ne L^{i_2}\mathfrak{u} + \mathfrak{s}$. Then by Equality (11), we have $(L^{k+i_1} + L^{i_2})\mathfrak{u} = (L^k + I)\mathfrak{s}$. It follows from Lemma 4 that there exists some integer k' with $0 \le k' < 2^n - 1$ such that

$$(L^{k+i_1} + L^{i_2})\mathfrak{u} = (L^k + I)\mathfrak{s} = L^{k'}\mathfrak{s}.$$

This is a contradiction of the fact that $\mathrm{per}((L^{k+i_1} + L^{i_2})\mathfrak{u}) = 3$ or 1 and $\mathrm{per}(L^{k'}\mathfrak{s}) = 2^n - 1$. The claim thus follows.

In the case of $\mathbf{a} = \overline{\mathfrak{u}}$ and $\mathbf{b} = \mathfrak{s}$, it can be similarly proved that the cycles $[\overline{\mathfrak{u}} + \mathfrak{s}]$, $[L\overline{\mathfrak{u}} + \mathfrak{s}]$, $[L^2\overline{\mathfrak{u}} + \mathfrak{s}]$ are the same for odd n, and pairwise distinct for even n.

On combining Cases 1 and 2, we deduce the decomposition of $\Omega((1 + x^3)p(x))$ given by Equalities (7) and (8). ∎

Remark 1: In Proposition 1, $\mathfrak{u}$ is one of the three sequences in the cycle $[(100)]$ and $\mathfrak{s}$ is one of the $2^n - 1$ m-sequences in $\Omega(p(x))$. Thus, the sequences $\mathfrak{u}$ and $\mathfrak{s}$ have 3 and $2^n - 1$ choices, respectively.

The following example illustrates the results of Proposition 1.

Example 2: (i) Let $n = 3$ and $p(x) = x^3 + x^2 + 1$. By Example 1, $\mathfrak{s} = (1101001)$ is an m-sequence of order 3 in $\Omega(p(x))$. Furthermore, the LFSR in $\mathcal{F}$ has characteristic polynomial $q(x) = (1 + x^3)(x^3 + x^2 + 1) = x^6 + x^5 + x^2 + 1$. Take $\mathfrak{u} = (100)$. Then the sum sequence $\mathfrak{u} + \mathfrak{s}$ is determined by

$$\begin{aligned}\mathfrak{u} &= \underline{100}100100100100100\ldots \\ \mathfrak{s} &= \underline{1101001}11010011101001\ldots \\ \mathfrak{u} + \mathfrak{s} &= \underline{0100000111101110011101}\ldots\end{aligned}$$

A short calculation shows that

$$\begin{aligned}&\Omega(q(x)) \\ &\quad = [\mathbf{0}] \cup [\mathbf{1}] \cup [(100)] \cup [(011)] \cup [(1101001)] \\ &\qquad \cup [(0010110)] \cup [(01000001111010111001101)] \\ &\qquad \cup [(10111110000010001100 10)].\end{aligned}$$

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(ii) Let $n = 4$ and $p(x) = x^4 + x^3 + 1$. Then $\mathfrak{s} = (110101100100011)$ is an m-sequence of order 4 in $\Omega(p(x))$. Furthermore, the LFSR in $\mathcal{F}$ has characteristic polynomial $q(x) = (1 + x^3)(x^4 + x^3 + 1) = x^7 + x^6 + x^4 + 1$. By setting $\mathfrak{u} = (100)$, a short calculation shows that

$$\begin{aligned}\Omega(q(x)) &= [\mathbf{0}] \cup [\mathbf{1}] \cup [(100)] \cup [(011)] \cup [(110101100100011)] \\ &\quad \cup [(001010011011100)] \cup [(010001000000111)] \\ &\quad \cup [(111100101101010)] \cup [(100111110110001)] \\ &\quad \cup [(101110111111000)] \cup [(000011010010101)] \\ &\quad \cup [(011000001001110)],\end{aligned}$$

where $[L^i\mathfrak{u} + \mathfrak{s}]$ and $[L^i\overline{\mathfrak{u}} + \mathfrak{s}]$ for $i = 0, 1, 2$ are calculated in a similar way as in Case (i).

We call $\overline{\mathbf{a}} = \mathbf{a} + \mathbf{1}$ the *complement* of the sequence $\mathbf{a}$. The cycles $[\mathbf{a}]$ and $[\overline{\mathbf{a}}]$ are said to be *complementary cycles*. In particular, $[\mathbf{a}]$ is called a *self-complementary cycle* if $[\mathbf{a}] = [\overline{\mathbf{a}}]$. From Proposition 1, we observe that $\Omega(q(x))$ consists of complementary cycles, namely, the cycles $[\mathbf{a}]$ and $[\overline{\mathbf{a}}]$ appear in pairs in $\Omega(q(x))$. Such an observation will considerably ease the investigation of the adjacency graphs of LFSRs in $\mathcal{F}$. We end this section by addressing a corollary from Lemma 3 and Proposition 1, which will be used in the sequel.

Corollary 1: With notation in Proposition 1, the decompositions of $\Omega(f(x))$ for some proper factors $f(x)$ of $q(x)$ are shown as follows.

(i) $\Omega(1 + x) = [\mathbf{0}] \cup [\mathbf{1}]$.
(ii) $\Omega(1 + x + x^2) = [\mathbf{0}] \cup [\overline{\mathfrak{u}}]$.
(iii) $\Omega(1 + x^3) = [\mathbf{0}] \cup [\mathbf{1}] \cup [\mathfrak{u}] \cup [\overline{\mathfrak{u}}]$.
(iv) $\Omega(p(x)) = [\mathbf{0}] \cup [\mathfrak{s}]$.
(v) $\Omega((1 + x)p(x)) = [\mathbf{0}] \cup [\mathbf{1}] \cup [\mathfrak{s}] \cup [\overline{\mathfrak{s}}]$.
(vi) $\Omega((1 + x + x^2)p(x)) = [\mathbf{0}] \cup [\mathfrak{s}] \cup [\overline{\mathfrak{u}}] \cup [\overline{\mathfrak{u}} + \mathfrak{s}]$ for odd n, and $\Omega((1+x+x^2)p(x)) = [\mathbf{0}] \cup [\mathfrak{s}] \cup [\overline{\mathfrak{u}}] \bigcup_{i=0}^{2} [L^i\overline{\mathfrak{u}} + \mathfrak{s}]$ for even n.

IV. Adjacency Graphs of LFSRs in $\mathcal{F}$

In this section, the adjacency graphs of LFSRs in $\mathcal{F}$ (defined in Section III) are determined.

For a given $(n + 3)$-stage LFSR in $\mathcal{F}$, its characteristic polynomial $q(x)$ has the form as in (5). To determine the adjacency graph, by Proposition 1, we need to determine the conjugate pairs shared by the cycles in $\Omega(q(x))$.

In the sequel, we will first show the nonexistence of conjugate pairs between two cycles or on one cycle in $\Omega(q(x))$, which enables us to find the adjacent cycle of each of the two cycles $[\mathbf{0}]$ and $[\mathbf{1}]$. As a consequence, some state equations are also obtained. By further applying the properties of m-sequences to these equations, the conjugate pairs between other cycles in $\Omega(q(x))$ will be determined.

The following lemma will be heavily used in investigating the adjacent cycles in $\Omega(q(x))$.

Lemma 5: Let $f(x)$ be a polynomial over $\mathbb{F}_2$ of degree n and $f_1(x)$ be a proper factor of $f(x)$.

(i) For a cycle $[\mathbf{a}] \in \Omega(f(x))$, if $[\mathbf{a}]$ also belongs to $\Omega(f_1(x))$, then $[\mathbf{a}]$ in $\Omega(f(x))$ doesn't contain any conjugate pair.
(ii) For two cycles $[\mathbf{a}]$, $[\mathbf{b}] \in \Omega(f(x))$, if $[\mathbf{a}]$ and $[\mathbf{b}]$ are also in $\Omega(f_1(x))$ and $[\mathbf{a}] \cap [\mathbf{b}] = \emptyset$, then no conjugate pair is shared by $[\mathbf{a}]$ and $[\mathbf{b}]$ in $\Omega(f(x))$.

Proof: See Appendix A. ∎

Applying Lemma 5 to $\Omega(q(x))$ immediately gives the following proposition.

Proposition 2: None of the cycles in $\Omega(q(x))$ contains a conjugate pair.

Proof: Consider the case that n is odd. By Corollary 1 (iii), (v), (vi) and Lemma 5, none of the seven cycles $[\mathbf{0}]$, $[\mathbf{1}]$, $[\mathfrak{u}]$, $[\overline{\mathfrak{u}}]$, $[\mathfrak{s}]$, $[\overline{\mathfrak{s}}]$, $[\overline{\mathfrak{u}} + \mathfrak{s}]$ contains a conjugate pair. Then by Proposition 1 (i), only one cycle $[\mathfrak{u}+\mathfrak{s}]$ remains to be considered. We claim that $[\mathfrak{u} + \mathfrak{s}]$ doesn't contain a conjugate pair. Suppose that $[\mathfrak{u} + \mathfrak{s}]$ contains a conjugate pair $\{\mathbf{A}, \widehat{\mathbf{A}}\}$. Since $[\mathfrak{u} + \mathfrak{s}]$ and $[\overline{\mathfrak{u}} + \mathfrak{s}]$ are complementary cycles, it follows that $[\overline{\mathfrak{u}}+\mathfrak{s}]$ contains the two states $\overline{\mathbf{A}}$ and $\overline{\widehat{\mathbf{A}}}$. By Equality (2), $\widehat{\overline{\mathbf{A}}} = \overline{\widehat{\mathbf{A}}}$. This implies that $[\overline{\mathfrak{u}} + \mathfrak{s}]$ contains a conjugate pair $\{\overline{\mathbf{A}}, \widehat{\overline{\mathbf{A}}}\}$, a contradiction. Thus, the claim and then the proposition hold for the case that n is odd.

The case that n is even can be proved similarly. It follows from Corollary 1 (iii), (v), (vi) and Lemma 5 that none of the nine cycles $[\mathbf{0}]$, $[\mathbf{1}]$, $[\mathfrak{u}]$, $[\overline{\mathfrak{u}}]$, $[\mathfrak{s}]$, $[\overline{\mathfrak{s}}]$, $[\overline{\mathfrak{u}}+\mathfrak{s}]$, $[L\overline{\mathfrak{u}}+\mathfrak{s}]$ and $[L^2\overline{\mathfrak{u}}+\mathfrak{s}]$ contains a conjugate pair. Then by Proposition 1 (ii), it suffices to consider the three cycles $[L^i\mathfrak{u}+\mathfrak{s}]$ for $i \in \{0, 1, 2\}$. Note that $[L^i\mathfrak{u}+\mathfrak{s}]$ and $[L^i\overline{\mathfrak{u}}+\mathfrak{s}]$ are complementary cycles for each $i \in \{0, 1, 2\}$. The remainder of the proof is completely analogous to the case that n is odd, so we omit it here. ∎

Note that if $\mathbf{A}$ is a state on a cycle $[\mathbf{a}]$ in $\Omega(q(x))$ and its conjugate $\widehat{\mathbf{A}}$ is on another cycle $[\mathbf{b}]$, then the two cycles $[\overline{\mathbf{a}}]$ and $[\overline{\mathbf{b}}]$ in $\Omega(q(x))$ contain $\overline{\mathbf{A}}$ and $\overline{\widehat{\mathbf{A}}}$, respectively. This together with the equality $\widehat{\overline{\mathbf{A}}} = \overline{\widehat{\mathbf{A}}}$ implies that the cycles $[\overline{\mathbf{a}}]$ and $[\overline{\mathbf{b}}]$ contain a conjugate pair $\{\overline{\mathbf{A}}, \widehat{\overline{\mathbf{A}}}\}$. We thus have the following lemma.

Lemma 6: If two distinct cycles $[\mathbf{a}]$ and $[\mathbf{b}]$ in $\Omega(q(x))$ contain exactly r conjugate pairs, then their complement cycles $[\overline{\mathbf{a}}]$ and $[\overline{\mathbf{b}}]$ also contain exactly r conjugate pairs.

Lemma 6 shows some "symmetry" properties of the adjacency graphs of the LFSRs in $\mathcal{F}$. This fact will ease our analysis of the adjacent cycles in $\Omega(q(x))$ in the sequel. Namely, if we aim to determine all possible adjacent cycles of two complementary cycles $[\mathbf{a}]$ and $[\overline{\mathbf{a}}]$, then it suffices to find the adjacent cycles of $[\mathbf{a}]$ and those of $[\overline{\mathbf{a}}]$ are obtained directly by Lemma 6.

Now we can give the adjacent cycles of two cycles $[\mathbf{0}]$ and $[\mathbf{1}]$.

Proposition 3: (i) If n is odd, then the cycle $[\mathbf{0}]$ is adjacent to the cycle $[\mathfrak{u} + \mathfrak{s}]$, and the cycle $[\mathbf{1}]$ is adjacent to the cycle $[\overline{\mathfrak{u}} + \mathfrak{s}]$.

(ii) If n is even, then the cycle $[\mathbf{0}]$ is adjacent to one of the three cycles $[\mathfrak{u} + \mathfrak{s}]$, $[L\mathfrak{u} + \mathfrak{s}]$ and $[L^2\mathfrak{u} + \mathfrak{s}]$. Similarly, the cycle $[\mathbf{1}]$ is adjacent to one of the three cycles $[\overline{\mathfrak{u}}+\mathfrak{s}]$, $[L\overline{\mathfrak{u}}+\mathfrak{s}]$ and $[L^2\overline{\mathfrak{u}} + \mathfrak{s}]$.

Proof: For odd n, by Corollary 1 (iii), (v), (vi) and Lemma 5, $[\mathbf{0}]$ doesn't share any conjugate pair with each of the six cycles $[\mathbf{1}]$, $[\mathfrak{u}]$, $[\overline{\mathfrak{u}}]$, $[\mathfrak{s}]$, $[\overline{\mathfrak{s}}]$ and $[\overline{\mathfrak{u}} + \mathfrak{s}]$. Note that $[\mathbf{0}]$

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contains exactly one state, i.e., the $(n+3)$-tuple $(0,0,\ldots,0)$. It follows from Proposition 1 (i) that the conjugate of $(0,0,\ldots,0)$, i.e., the $(n+3)$-tuple $(1,0,\ldots,0)$ must be on the cycle $[\mathfrak{u}+\mathfrak{s}]$. That is, the cycle $[\mathbf{0}]$ is adjacent to the cycle $[\mathfrak{u}+\mathfrak{s}]$. Since $[\mathbf{0}]$ and $[\mathbf{1}]$ are complementary cycles, by Lemma 6, the cycle $[\mathbf{1}]$ is adjacent to the complementary cycle of $[\mathfrak{u}+\mathfrak{s}]$, i.e., the cycle $[\overline{\mathfrak{u}}+\mathfrak{s}]$.

For even n, according to Corollary 1 (iii), (v), (vi) and Lemma 5, $[\mathbf{0}]$ doesn't share any conjugate pair with each of the eight cycles $[\mathbf{1}]$, $[\mathfrak{u}]$, $[\overline{\mathfrak{u}}]$, $[\mathfrak{s}]$, $[\overline{\mathfrak{s}}]$, $[\overline{\mathfrak{u}}+\mathfrak{s}]$, $[L\overline{\mathfrak{u}}+\mathfrak{s}]$ and $[L^2\overline{\mathfrak{u}}+\mathfrak{s}]$. Then by Proposition 1 (ii), the $(n+3)$-tuple $(1,0,\ldots,0)$ must be on one of $[\mathfrak{u}+\mathfrak{s}]$, $[L\mathfrak{u}+\mathfrak{s}]$ and $[L^2\mathfrak{u}+\mathfrak{s}]$. Note that $[L^i\mathfrak{u}+\mathfrak{s}]$ and $[L^i\overline{\mathfrak{u}}+\mathfrak{s}]$ are complementary cycles for each $i\in\{0,1,2\}$. Therefore, the second statement follows from Lemma 6. ∎

Recall that Equalities (7) and (8) hold for any $\mathfrak{u}\in[(100)]$ and any nonzero $\mathfrak{s}\in\Omega(p(x))$. Let $\mathfrak{u}=(u_0u_1u_2)\in[(100)]$ and $\mathfrak{s}=(s_0s_1\ldots s_{2^n-2})\in\Omega(p(x))$. Define the $(n+3)$-tuples

$$\mathcal{U}_i=(u_i,u_{i+1},\ldots,u_{i+n+2}),\ \mathcal{S}_j=(s_j,s_{j+1},\ldots,s_{j+n+2}),$$

where $0\le i\le 2$, $0\le j\le 2^n-2$, and for each $k\ge 0$, the subscripts of u_{i+k} and s_{j+k} are taken modulo 3 and 2^n-1 respectively. Then the sequences $\mathfrak{u}$ and $\mathfrak{s}$ can be rewritten in terms of states (as the notation in Section II-B):

$$\mathfrak{u}=(\mathcal{U}_0\mathcal{U}_1\mathcal{U}_2),\quad \mathfrak{s}=(\mathcal{S}_0\mathcal{S}_1\ldots\mathcal{S}_{2^n-2}),$$

where the subscripts of $\mathcal{U}$ and $\mathcal{S}$ are taken modulo 3 and 2^n-1 respectively. Let $\mathbf{E}$ denote the $(n+3)$-tuple $(1,0,\ldots,0)$.

With the notation given in the last paragraph, for odd n, every state on the cycle $[\mathfrak{u}+\mathfrak{s}]$ has the form as $\mathcal{U}_k+\mathcal{S}_k$, where $0\le k\le 3(2^n-1)-1$. Then by Proposition 3 (i), $\mathbf{E}$ is on the cycle $[\mathfrak{u}+\mathfrak{s}]$, which implies $\mathcal{U}_{k_0}+\mathcal{S}_{k_0}=\mathbf{E}$ for some integer $k_0\in\{0,1,\ldots,2^n-2\}$. In this case, we define $\mathbf{u}=(\mathbf{U}_0\mathbf{U}_1\mathbf{U}_2)=L^{k_0}\mathfrak{u}$ and $\mathbf{s}=(\mathbf{S}_0\mathbf{S}_1\ldots\mathbf{S}_{2^n-2})=L^{k_0}\mathfrak{s}$. Then we have $\mathbf{U}_k=\mathcal{U}_{k_0+k}$ and $\mathbf{S}_k=\mathcal{S}_{k_0+k}$ for $0\le k\le 3(2^n-1)-1$ and

$$\mathbf{U}_0+\mathbf{S}_0=\mathcal{U}_{k_0}+\mathcal{S}_{k_0}=\mathbf{E}.$$

If n is even, then every state on the cycle $[L^i\mathfrak{u}+\mathfrak{s}]$, where $i\in\{0,1,2\}$, has the form as $\mathcal{U}_{i+k}+\mathcal{S}_k$, where $0\le k\le 2^n-2$. It follows from Proposition 3 (ii) that $\mathbf{E}$ is on a cycle $[L^{i_0}\mathfrak{u}+\mathfrak{s}]$, where $i_0\in\{0,1,2\}$. This implies $\mathcal{U}_{i_0+k_0}+\mathcal{S}_{k_0}=\mathbf{E}$ for some integer $k_0\in\{0,1,\ldots,2^n-2\}$. Similarly, we define $\mathbf{u}=(\mathbf{U}_0\mathbf{U}_1\mathbf{U}_2)=L^{i_0+k_0}\mathfrak{u}$ and $\mathbf{s}=(\mathbf{S}_0\mathbf{S}_1\ldots\mathbf{S}_{2^n-2})=L^{k_0}\mathfrak{s}$. Then, $\mathbf{U}_k=\mathcal{U}_{i_0+k_0+k}$ and $\mathbf{S}_k=\mathcal{S}_{k_0+k}$ for $0\le k\le 2^n-2$ and

$$\mathbf{U}_0+\mathbf{S}_0=\mathcal{U}_{i_0+k_0}+\mathcal{S}_{k_0}=\mathbf{E}.$$

Therefore, the sequences $\mathbf{u}=(\mathbf{U}_0\mathbf{U}_1\mathbf{U}_2)$ and $\mathbf{s}=(\mathbf{S}_0\mathbf{S}_1\ldots\mathbf{S}_{2^n-2})$ satisfy $\mathbf{U}_0+\mathbf{S}_0=\mathbf{E}$ regardless of whether n is even or odd.

Example 3: Let $\mathfrak{u}$ and $\mathfrak{s}$ be given as in Example 2. (i) For $n=3$, one has $\mathfrak{u}=(u_0u_1u_2)=(100)$, $\mathfrak{s}=(s_0s_1\ldots s_6)=(1101001)$ and $\mathfrak{u}+\mathfrak{s}=(01000001111101110011101)$. This implies $\mathcal{U}_1+\mathcal{S}_1=(1,0,0,0,0,0)=\mathbf{E}$. In this case, $\mathbf{U}_0=\mathcal{U}_1=(0,0,1,0,0,1)$ and $\mathbf{S}_0=\mathcal{S}_1=(1,0,1,0,0,1)$. (ii) For $n=4$, one has $\mathfrak{u}=(100)$, $\mathfrak{s}=(110101100100011)$ and $\mathfrak{u}+\mathfrak{s}=(010001000000111)$. This implies $\mathcal{U}_2+\mathcal{S}_5=(1,0,0,0,0,0,0)=\mathbf{E}$. In this case, $\mathbf{U}_0=\mathcal{U}_2=(0,1,0,0,1,0,0)$ and $\mathbf{S}_0=\mathcal{S}_5=(1,1,0,0,1,0,0)$.

For (i) and (ii), the sequences $\mathbf{u}=(\mathbf{U}_0\mathbf{U}_1\mathbf{U}_2)$ and $\mathbf{s}=(\mathbf{S}_0\mathbf{S}_1\ldots\mathbf{S}_{2^n-2})$ satisfy $\mathbf{U}_0+\mathbf{S}_0=\mathbf{E}$.

Note that $[\mathbf{u}]=[\mathfrak{u}]$ and $[\mathbf{s}]=[\mathfrak{s}]$. Throughout what follows, we shall adopt $[\mathbf{u}]$ and $[\mathbf{s}]$ to denote the cycles $[\mathfrak{u}]$ and $[\mathfrak{s}]$. It follows that $\mathbf{E}$ is always on the cycle $[\mathbf{u}+\mathbf{s}]$ and

$$\mathbf{U}_0+\mathbf{S}_0=\mathbf{E}. \tag{12}$$

In addition, it follows from $\mathbf{u}=(\mathbf{U}_0\mathbf{U}_1\mathbf{U}_2)\in[(100)]$ that

$$\mathbf{U}_0+\mathbf{U}_1+\mathbf{U}_2=\mathbf{1}_{n+3}, \tag{13}$$

where $\mathbf{1}_{n+3}$ is the all-one $(n+3)$-tuple.

For the sequence $\mathbf{s}=(\mathbf{S}_0\mathbf{S}_1\ldots\mathbf{S}_{2^n-2})$, where $\mathbf{S}_j=(s_j,s_{j+1},\ldots,s_{j+n+2})$ and $0\le j\le 2^n-2$, define

$$\widetilde{\mathbf{S}}_j=(s_j,s_{j+1},\ldots,s_{j+n-1}), \tag{14}$$

where the subscript of s_{j+k} is taken modulo 2^n-1 for each $k\ge 0$. In the following lemma we describe some equations of the $(n+3)$-tuples $\mathbf{S}_j$, $0\le j\le 2^n-2$, derived from the m-sequence $\mathbf{s}$.

Lemma 7: For an integer j in the set $J^*=\{1,2,\ldots,2^n-2\}$, there exists an integer $k\in J^*$ such that $\widetilde{\mathbf{S}}_0+\widetilde{\mathbf{S}}_j=\widetilde{\mathbf{S}}_k$. Moreover, the equality $\mathbf{S}_0+\mathbf{S}_j=\mathbf{S}_k$ holds.

Proof: See Appendix A. ∎

Lemma 7 motivates us to derive some equations between states $\mathbf{S}_0$ and $\mathbf{S}_j$ from those between states $\widetilde{\mathbf{S}}_0$ and $\widetilde{\mathbf{S}}_j$, where $j\in J^*$. In the notation of Lemma 7, we have $\widetilde{\mathbf{S}}_{i+1}=(s_{i+1},s_{i+2},\ldots,s_{i+n-1},\sum_{t=0}^{n-1}c_ts_{i+t})$ for $i\ge 0$. This can be rewritten as $\widetilde{\mathbf{S}}_{i+1}=\widetilde{\mathbf{S}}_iM$, where M is an $n\times n$ matrix over $\mathbb{F}_2$ defined by

$$M=\begin{pmatrix}0&0&0&\ldots&0&0&c_0\\1&0&0&\ldots&0&0&c_1\\0&1&0&\ldots&0&0&c_2\\0&0&1&\ldots&0&0&c_3\\\vdots&\vdots&\vdots&\ddots&\vdots&\vdots&\vdots\\0&0&0&\ldots&1&0&c_{n-2}\\0&0&0&\ldots&0&1&c_{n-1}\end{pmatrix}.$$

It follows that

$$\widetilde{\mathbf{S}}_i=\widetilde{\mathbf{S}}_{i-1}M=\cdots=\widetilde{\mathbf{S}}_1M^{i-1}=\widetilde{\mathbf{S}}_0M^i \tag{15}$$

for all $i\ge 0$.

Note that M is the companion matrix of $p(x)$, then $p(M)=\mathbb{O}$, where $\mathbb{O}$ is the $n\times n$ zero matrix [21]. Since $p(x)$ is a primitive polynomial of degree n over $\mathbb{F}_2$, the ring $\mathbb{F}_2[M]$ is isomorphic to the field $\mathbb{F}_{2^n}$ and M is a generator of the multiplicative group of $\mathbb{F}_2[M]$ [25], [34]. Let α be a primitive element of $\mathbb{F}_{2^n}$. Define the isomorphism mapping $\phi:\mathbb{F}_{2^n}\to\mathbb{F}_2[M]$ by

$$\phi(0)=\mathbb{O},\ \ \phi(\alpha^j)=M^j, \tag{16}$$

where $j\in J^*$. This isomorphism gives

$$I_n+M^j=\phi(1)+\phi(\alpha^j)=\phi(1+\alpha^j)=\phi(\alpha^{Z(j)})=M^{Z(j)},$$

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where $1+\alpha^i=\alpha^{Z(j)}$ comes from Equality (4). Furthermore, by Equality (15),

$$\widetilde{\mathbf{S}}_0+\widetilde{\mathbf{S}}_j=\widetilde{\mathbf{S}}_0(I_n+M^j)=\widetilde{\mathbf{S}}_0M^{Z(j)}=\widetilde{\mathbf{S}}_{Z(j)}$$

for any $j\in J^*$. It then follows from Lemma 7 that for any $j\in J^*$,

$$\mathbf{S}_0+\mathbf{S}_j=\mathbf{S}_{Z(j)}. \tag{17}$$

By Equalities (12) and (17), for any $j\in J^*$, we have

$$\mathbf{U}_0+\mathbf{S}_j=\mathbf{E}+\mathbf{S}_{Z(j)}=\widehat{\mathbf{S}}_{Z(j)}. \tag{18}$$

This equality together with the permutation Z over the set J^* motivates our approach to investigating the conjugate pairs between the cycles in $\Omega(q(x))$. By Proposition 1, our investigation is split into two cases, n is odd and n is even.

A. The Case That n is Odd

In this subsection, we determine the adjacency graph of any $(n+3)$-stage LFSR in $\mathcal{F}$ for odd n. Since the adjacent cycle of the cycle $[\mathbf{0}]$ and that of $[\mathbf{1}]$ are given by Proposition 3 (i), by Proposition 1 (i), it remains to find the conjugate pairs between any two of the other six cycles in $\Omega(q(x))$.

In the following, by the permutation Z, Equalities (12) and (17), the conjugates of the states on the two cycles $[\mathbf{s}]$ and $[\overline{\mathbf{s}}]$ will first be determined. Similarly to Proposition 3, the conjugate pairs shared by any two of the four cycles $[\overline{\mathbf{u}}]$, $[\mathbf{u}+\mathbf{s}]$, $[\overline{\mathbf{u}}+\mathbf{s}]$ and $[\mathbf{u}]$ can also be obtained. Thus, the adjacency graph is determined.

To proceed with our analysis, let Δ be the set of the states on the cycle $[\mathbf{u}+\mathbf{s}]$, then by Proposition 1 we have

$$\Delta=\left\{\mathbf{U}_i+\mathbf{S}_i\,|\,0\le i<3(2^n-1)\right\},$$

where the subscripts of $\mathbf{U}_i$ are taken modulo 3 and those of $\mathbf{S}_i$ are taken modulo 2^n-1. Denote a subset of Δ by $\Delta_1=\left\{\mathbf{U}_{3j}+\mathbf{S}_{3j}\,|\,0\le j<2^n-1\right\}$. Since $\gcd(3,2^n-1)=1$, it follows that $3j \pmod{2^n-1}$ runs through the set J when j runs through the set J. Note that $\mathbf{U}_{3j}=\mathbf{U}_0$, then

$$\Delta_1=\left\{\mathbf{U}_0+\mathbf{S}_j\,|\,j\in J\right\}. \tag{19}$$

This together with Equalities (12) and (18) implies that

$$\begin{aligned}\Delta_1&=\{\mathbf{U}_0+\mathbf{S}_0\}\bigcup\left\{\mathbf{U}_0+\mathbf{S}_j\,|\,j\in J^*\right\}\\&=\{\mathbf{E}\}\bigcup\left\{\widehat{\mathbf{S}}_{Z(j)}\,|\,j\in J^*\right\}\\&=\{\mathbf{E}\}\bigcup\left\{\widehat{\mathbf{S}}_j\,|\,j\in J^*\right\},\end{aligned} \tag{20}$$

the last equality holds due to the fact that the mapping Z is a permutation of the set J^*. Note that Δ_1 consists of states on the cycle $[\mathbf{u}+\mathbf{s}]$ and $\mathbf{S}_j$ is on the cycle $[\mathbf{s}]$ for any $j\in J^*$. Then we obtain the adjacent cycles of two cycles $[\mathbf{s}]$ and $[\overline{\mathbf{s}}]$ in $\Omega(q(x))$.

Proposition 4: If n is odd, then the cycle $[\mathbf{s}]$ shares exactly 2^n-2 conjugate pairs with the cycle $[\mathbf{u}+\mathbf{s}]$ and one conjugate pair with the cycle $[\mathbf{u}]$. The cycle $[\overline{\mathbf{s}}]$ shares exactly 2^n-2 conjugate pairs with the cycle $[\overline{\mathbf{u}}+\mathbf{s}]$ and one conjugate pair with the cycle $[\overline{\mathbf{u}}]$.

Proof: By Lemma 6, it suffices to show the case of the cycle $[\mathbf{s}]$. Actually, it follows from Equality (20) that $[\mathbf{s}]$ shares exactly 2^n-2 conjugate pairs with the cycle $[\mathbf{u}+\mathbf{s}]$. Note that $\mathbf{S}_0=\mathbf{E}+\mathbf{U}_0=\widehat{\mathbf{U}}_0$ by Equality (12). Then $[\mathbf{s}]$ shares exactly one conjugate pair with the cycle $[\mathbf{u}]$. Since $[\mathbf{s}]$ contains exactly 2^n-1 states, the adjacent cycles of the cycle $[\mathbf{s}]$ are completely determined. ■

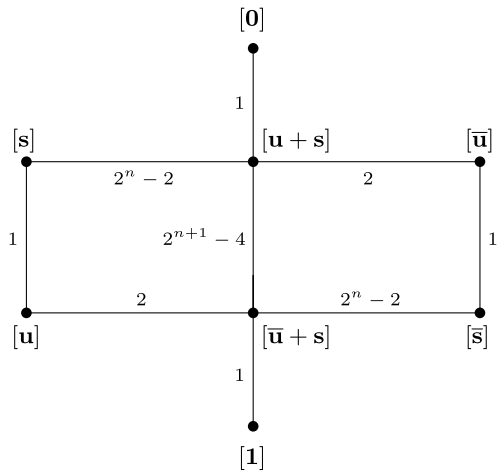


Fig. 2. The adjacency graph of an $(n+3)$-stage LFSR in $\mathcal{F}$.

We are now ready to give the adjacency graph of any one $(n+3)$-stage LFSR in $\mathcal{F}$ in the case that n is odd.

Theorem 1: If n is odd, then the adjacency graph of an $(n+3)$-stage LFSR in $\mathcal{F}$ is shown in Fig. 2.

Proof: By Propositions 3 and 4, the conjugates of states on the four cycles $[\mathbf{0}]$, $[\mathbf{1}]$, $[\mathbf{s}]$ and $[\overline{\mathbf{s}}]$ are completely determined. As a consequence, six edges with labeled numbers in the adjacency graph are obtained.

To complete the adjacency graph, by Propositions 1 (i) and 2, it remains to determine the conjugate pairs shared by any two of the four cycles $[\overline{\mathbf{u}}]$, $[\mathbf{u}+\mathbf{s}]$, $[\overline{\mathbf{u}}+\mathbf{s}]$ and $[\mathbf{u}]$.

We claim that the two cycles $[\overline{\mathbf{u}}]$ and $[\mathbf{u}+\mathbf{s}]$ share exactly two conjugate pairs. The proof of the claim is analogous to that of Proposition 3. By Corollary 1 (iii), (vi) and Lemma 5, $[\overline{\mathbf{u}}]$ doesn't share any conjugate pair with cycles $[\mathbf{u}]$ and $[\overline{\mathbf{u}}+\mathbf{s}]$. Note that $[\overline{\mathbf{u}}]$ contains exactly three states, and by Proposition 4, $[\overline{\mathbf{u}}]$ shares exactly one conjugate pair with $[\overline{\mathbf{s}}]$. It follows from Proposition 2 that there are exactly two conjugate pairs between the two cycles $[\overline{\mathbf{u}}]$ and $[\mathbf{u}+\mathbf{s}]$. Thus, the claim is proved. By the claim and Lemma 6, there are exactly two conjugate pairs between the two cycles $[\mathbf{u}]$ and $[\overline{\mathbf{u}}+\mathbf{s}]$.

Now it suffices to consider the conjugate pairs shared by the two cycles $[\mathbf{u}+\mathbf{s}]$ and $[\overline{\mathbf{u}}+\mathbf{s}]$. Consequently, by Propositions 1 (i) and 2, there exist exactly $3(2^n-1)-(2^n-2+1+2)=2^{n+1}-4$ conjugate pairs between the two cycles $[\mathbf{u}+\mathbf{s}]$ and $[\overline{\mathbf{u}}+\mathbf{s}]$. ■

B. The Case That n is Even

In this subsection, for even n, the adjacency graph of any $(n+3)$-stage LFSR in $\mathcal{F}$ will be determined. Since the adjacent cycle of the cycle $[\mathbf{0}]$ and that of $[\mathbf{1}]$ are characterized by Proposition 3 (ii), by Proposition 1 (ii), it remains to find the conjugate pairs between any two of the other ten cycles in $\Omega(q(x))$.

First, we shall employ Equality (12) to analyze some special states on the cycle $[L^i\mathbf{u}+\mathbf{s}]$ for $i\in\{0,1,2\}$, and obtain

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the conjugates of states on the two cycles $[\mathbf{u}]$ and $[\overline{\mathbf{u}}]$. Then by Lemma 1 and Equality (17), the conjugate pairs shared by any two of the remaining eight cycles are considered. Consequently, the adjacency graph is determined.

Before proceeding we introduce some notation. For $i \in \{0, 1, 2\}$, let Φ_i be the set of states on the cycle $[L^i\mathbf{u} + \mathbf{s}]$. Then by Proposition 1 and Equality (3),

$$\Phi_i = \left\{\mathbf{U}_{i+j} + \mathbf{S}_j \mid j \in J\right\}, \tag{21}$$

where the subscripts of $\mathbf{U}_{i+j}$ are taken modulo 3 and those of $\mathbf{S}_j$ are taken modulo $2^n - 1$. For each $i \in \{0, 1, 2\}$, define two sets $\overline{\Phi}_i$ and $\widehat{\Phi}_i$ by

$$\overline{\Phi}_i = \left\{\overline{\mathbf{A}} \mid \mathbf{A} \in \Phi_i\right\}, \quad \widehat{\Phi}_i = \left\{\widehat{\mathbf{A}} \mid \mathbf{A} \in \Phi_i\right\}. \tag{22}$$

For any subset $\Phi' \subset \Phi_i$, the sets $\overline{\Phi}'$ and $\widehat{\Phi}'$ can be defined in the same manner. Obviously, $\overline{\Phi}_i$ is the set of states on the cycle $[L^i\overline{\mathbf{u}} + \mathbf{s}]$ for each $i \in \{0, 1, 2\}$.

For each $i \in \{0, 1, 2\}$, we have $\Phi_i = \{\mathbf{U}_i + \mathbf{S}_0\} \cup \Phi_i^*$, where Φ_i^* is defined by

$$\Phi_i^* = \left\{\mathbf{U}_{i+j} + \mathbf{S}_j \mid j \in J^*\right\}. \tag{23}$$

Consider the state $\mathbf{U}_i + \mathbf{S}_0$ in Φ_i for each $i \in \{0, 1, 2\}$. By Proposition 1 and Equality (13), for each $l \in \{1, 2\}$ we have $\mathbf{U}_l + \mathbf{U}_0 = \overline{\mathbf{U}}_{3-l}$. Then by Equality (12) we have

$$\mathbf{U}_i + \mathbf{S}_0 = \begin{cases} \mathbf{U}_0 + \mathbf{S}_0 = \mathbf{E}, & i = 0, \\ \mathbf{U}_1 + \mathbf{S}_0 = \mathbf{U}_0 + \overline{\mathbf{U}}_2 + \mathbf{S}_0 = \mathbf{E} + \overline{\mathbf{U}}_2, & i = 1, \\ \mathbf{U}_2 + \mathbf{S}_0 = \mathbf{U}_0 + \overline{\mathbf{U}}_1 + \mathbf{S}_0 = \mathbf{E} + \overline{\mathbf{U}}_1, & i = 2. \end{cases} \tag{24}$$

Note that $\mathbf{U}_i$ and $\overline{\mathbf{U}}_i$ are on the cycles $[\mathbf{u}]$ and $[\overline{\mathbf{u}}]$, respectively. By Equality (24), the case $i = 0$ implies that $\mathbf{U}_0 = \mathbf{S}_0 + \mathbf{E} = \widehat{\mathbf{S}}_0$. It follows that the cycle $[\mathbf{u}]$ shares a conjugate pair with the cycle $[\mathbf{s}]$. By Lemma 6, the cycle $[\overline{\mathbf{u}}]$ shares a conjugate pair with the cycle $[\overline{\mathbf{s}}]$.

By Equality (24), the case $i = 1$ yields that $\mathbf{U}_1 + \mathbf{S}_0 = \widehat{\overline{\mathbf{U}}}_2$. Since $\mathbf{U}_1 + \mathbf{S}_0$ is on the cycle $[L\mathbf{u} + \mathbf{s}]$, it follows that $[L\mathbf{u} + \mathbf{s}]$ shares a conjugate pair with the cycle $[\overline{\mathbf{u}}]$. It can be similarly proved that $[L^2\mathbf{u} + \mathbf{s}]$ shares a conjugate pair with the cycle $[\overline{\mathbf{u}}]$. Note that the cycle $[\overline{\mathbf{u}}]$ contains exactly three states. Therefore, $[\overline{\mathbf{u}}]$ shares exactly one conjugate pair with each of the three cycles $[\overline{\mathbf{s}}]$, $[L\mathbf{u} + \mathbf{s}]$ and $[L^2\mathbf{u} + \mathbf{s}]$. Thus, by Lemma 6, the adjacent cycles of the two cycles $[\mathbf{u}]$ and $[\overline{\mathbf{u}}]$ are given in the following lemma.

Proposition 5: If n is even, then the cycle $[\mathbf{u}]$ shares exactly one conjugate pair with each of the three cycles $[\mathbf{s}]$, $[L\overline{\mathbf{u}} + \mathbf{s}]$ and $[L^2\overline{\mathbf{u}} + \mathbf{s}]$. The cycle $[\overline{\mathbf{u}}]$ shares exactly one conjugate pair with each of the three cycles $[\overline{\mathbf{s}}]$, $[L\mathbf{u} + \mathbf{s}]$ and $[L^2\mathbf{u} + \mathbf{s}]$.

For each $i \in \{0, 1, 2\}$, the conjugate of the state $\mathbf{U}_i + \mathbf{S}_0$ on the cycle $[L^i\mathbf{u} + \mathbf{s}]$ is determined by Equality (24). Therefore, in the sequel, we shall be mainly concerned with the set Φ_i^* instead of Φ_i. Let

$$\Phi_{i,i'}^* = \left\{\mathbf{U}_{i+i'} + \mathbf{S}_j \mid j \equiv i' \,(\mathrm{mod}\, 3),\ j \in J^*\right\}, \tag{25}$$

where $i, i' \in \{0, 1, 2\}$. Then $\Phi_i^* = \bigcup_{i'=0}^{2} \Phi_{i,i'}^*$ for each $i \in \{0, 1, 2\}$ with $\Phi_{i,i_1'}^* \cap \Phi_{i,i_2'}^* = \emptyset$ if $i_1' \neq i_2'$.

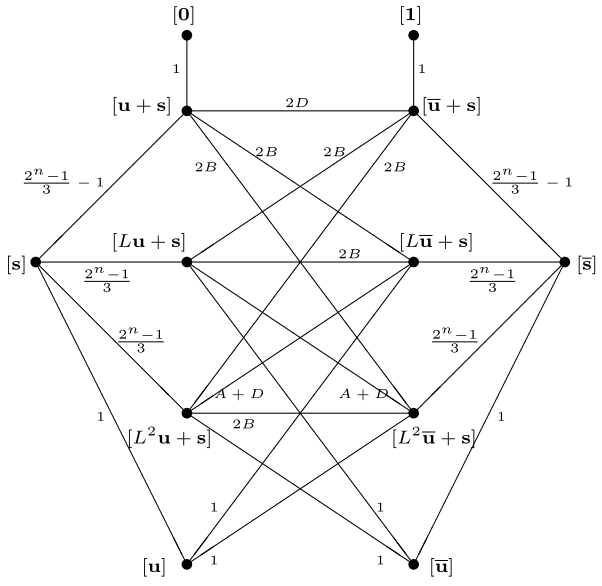


Fig. 3. The adjacency graph of an $(n + 3)$-stage LFSR in $\mathcal{F}$.

By Lemma 1, for each pair (x, y) where $x, y \in \{0, 1, 2\}$, the number of conjugate pairs shared by two cycles $[L^x\mathbf{u} + \mathbf{s}]$ and $[L^y\overline{\mathbf{u}} + \mathbf{s}]$ can be obtained similarly to Proposition 4.

Proposition 6: Let $N(x, y)$ be the number of conjugate pairs shared by two cycles $[L^x\mathbf{u} + \mathbf{s}]$ and $[L^y\overline{\mathbf{u}} + \mathbf{s}]$, where $x, y \in \{0, 1, 2\}$. Then $N(x, y) = (2-x, 1-y)_3 + (1-x, 2-y)_3$, where the addition between cyclotomic numbers is integer addition.

Proof: See Appendix A. ∎

The conjugates of the states on the cycle $[\mathbf{s}]$ follows from the proof of Proposition 6. Then, by Lemma 6, the conjugates of some states on the two cycles $[\mathbf{s}]$ and $[\overline{\mathbf{s}}]$ are given as follows.

Corollary 2: For even n, the cycle $[\mathbf{s}]$ shares exactly $\frac{2^n-1}{3} - 1$ conjugate pairs with the cycle $[\mathbf{u} + \mathbf{s}]$ and $\frac{2^n-1}{3}$ conjugate pairs with each of the cycles $[L\mathbf{u} + \mathbf{s}]$ and $[L^2\mathbf{u} + \mathbf{s}]$. The cycle $[\overline{\mathbf{s}}]$ shares exactly $\frac{2^n-1}{3} - 1$ conjugate pairs with the cycle $[\overline{\mathbf{u}} + \mathbf{s}]$ and $\frac{2^n-1}{3}$ conjugate pairs with each of the cycles $[L\overline{\mathbf{u}} + \mathbf{s}]$ and $[L^2\overline{\mathbf{u}} + \mathbf{s}]$.

Proof: See Appendix A. ∎

We can now give the adjacency graph of any one $(n + 3)$-stage LFSR in $\mathcal{F}$ in the case that n is even.

Theorem 2: If n is even, then the adjacency graph of an $(n+3)$-stage LFSR in $\mathcal{F}$ is shown in Fig. 3, where the numbers A, B and D are given by Lemma 2.

Proof: The conjugates of the states on the six cycles $[\mathbf{0}]$, $[\mathbf{1}]$, $[\mathbf{u}]$, $[\overline{\mathbf{u}}]$, $[\mathbf{s}]$ and $[\overline{\mathbf{s}}]$ are completely given by Propositions 3, 5 and Corollary 2. As a consequence, eleven edges labeled with numbers in the adjacency graph are obtained.

To complete the adjacency graph, by Proposition 1 (ii), it suffices to determine the conjugate pairs shared by the other six cycles in $\Omega(q(x))$, i.e., $[L^x\mathbf{u} + \mathbf{s}]$ and $[L^y\overline{\mathbf{u}} + \mathbf{s}]$ for $x, y \in \{0, 1, 2\}$.

By Corollary 1 (vi) and Lemma 5, any two of the three cycles $[\overline{\mathbf{u}} + \mathbf{s}]$, $[L\overline{\mathbf{u}} + \mathbf{s}]$ and $[L^2\overline{\mathbf{u}} + \mathbf{s}]$ are not adjacent. It then follows from Lemma 6 that the cycles $[\mathbf{u} + \mathbf{s}]$, $[L\mathbf{u} + \mathbf{s}]$ and $[L^2\mathbf{u} + \mathbf{s}]$ are pairwise nonadjacent, either. Therefore, for each $y \in \{0, 1, 2\}$, the cycle $[L^y\overline{\mathbf{u}} + \mathbf{s}]$ shares conjugate

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pairs only with the cycles $[L^x\mathbf{u}+\mathbf{s}]$ for all $x \in \{0,1,2\}$, and vice versa. Moreover, the number of conjugate pairs shared by the two cycles $[L^x\mathbf{u}+\mathbf{s}]$ and $[L^y\overline{\mathbf{u}}+\mathbf{s}]$ is given by Proposition 6 and Lemma 1. For instance, the number of conjugate pairs shared by the two cycles $[\mathbf{u}+\mathbf{s}]$ and $[\overline{\mathbf{u}}+\mathbf{s}]$ is given by $N(0,0)$, where $N(0,0)=(2,1)_3+(1,2)_3=2D$. The remaining cases can be determined similarly. ■

Remark 2: For fixed n, it follows from the definition in Section III that the $(n+3)$-stage LFSRs in $\mathcal{F}$ can be defined by different primitive polynomials. Note that the companion matrix of the primitive polynomial plays the role of a primitive element of the finite field $\mathbb{F}_{2^n}$ in Lemma 1 and it is uniquely determined by the primitive polynomial. Since the cyclotomic numbers are independent of the choice of primitive elements of the finite field, all the $(n+3)$-stage LFSRs in $\mathcal{F}$ have the same adjacency graph shown in Fig. 3.

The adjacency graphs of LFSRs in $\mathcal{F}$ are completely determined by Theorems 1 and 2. Let $p(x)$ be a primitive polynomial of degree n over $\mathbb{F}_2$ with $n>2$. The adjacency graphs of LFSRs with characteristic polynomial $(1+x)p(x)$ were essentially given in [29]. Hemmati determined the adjacency graphs of LFSRs with characteristic polynomial $(1+x)^2p(x)$ in [20]. Further, by recursively studying the properties of LFSRs with characteristic polynomial $(1+x)^m p(x)$ for any positive integer m, the adjacency graphs of LFSRs for $m=3$ were determined in [24]. The fact that $(1+x)^2p(x)=(1+x^2)p(x)$ motivates us to study the LFSRs with characteristic polynomial $(1+x^3)p(x)$. However, the recursive methods used in these papers don't seem to work for the case $(1+x^3)p(x)$ discussed in this paper.

In this section, by Lemma 7, Equalities (15) and (16), we establish a correspondence between the states $\mathbf{S}_j$, $0 \le j \le 2^n-2$, on the cycle $[\mathbf{s}]$ in $\Omega((1+x^3)p(x))$ and the elements in the multiplicative group $\mathbb{F}_{2^n}^*$. This correspondence enables us to derive equations $\mathbf{S}_0+\mathbf{S}_j=\mathbf{S}_{Z(j)}$ from $1+\alpha^j=\alpha^{Z(j)}$ for $j=1,2,\ldots,2^n-2$. Furthermore, with the connection between the permutation Z and the cyclotomic number described in Lemma 1, we are able to determine the number of conjugate pairs shared by two cycles $[L^x\mathbf{u}+\mathbf{s}]$ and $[L^y\overline{\mathbf{u}}+\mathbf{s}]$ for each pair (x,y) where $x,y\in\{0,1,2\}$.

V. Constructions of de Bruijn Sequences

In this section, we count the number of the de Bruijn sequences obtained by applying the cycle joining method to the LFSRs in $\mathcal{F}$. Moreover, we propose an algorithm for constructing $O(2^n)$ $(n+3)$-stage maximum-length NFSRs, where each NFSR is generated with time and memory complexity $O(n)$. At the end of this section, we sketch an approach to generating all the de Bruijn sequences, and the complexity is also discussed.

For a fixed LFSR in $\mathcal{F}$, its cycle structure and adjacency graph are completely determined by Proposition 1, Theorems 1 and 2. By employing the cycle joining method, one can join all the disjoint cycles generated by the LFSR. Thus, some de Bruijn sequences are constructed by this method since one can choose different conjugate pairs to join the cycles. To count the number of de Bruijn sequences constructed by the cycle joining method, we introduce the following lemmas.

Lemma 8: The number of spanning trees of the adjacency graph of an $(n+3)$-stage LFSR in $\mathcal{F}$ is $O(2^{3n})$ for odd n and $O(2^{7n})$ for even n.

Proof: By the well-known BEST Theorem [3] and Theorems 1 and 2, the exact number of spanning trees of the adjacency graph in Fig. 2 or Fig. 3 can be easily obtained. The desired conclusion then follows. ■

Lemma 9 ([22]): For $m\ge 4$, if we apply the cycle joining method to two different m-stage LFSRs, the resulting de Bruijn sequences are different.

By Fact 1, Lemmas 8 and 9, we have the following theorem.

Theorem 3: For a positive integer n with $n>2$, let Ψ_{n+3} be the set of the de Bruijn sequences of order $(n+3)$ obtained by applying the cycle joining method to all the $(n+3)$-stage LFSRs in $\mathcal{F}$. Then the size of Ψ_{n+3} is $O(2^{4n})$ for odd n and $O(2^{8n})$ for even n.

Proof: Given an $(n+3)$-stage LFSR in $\mathcal{F}$ with $n>2$, it follows from Fact 1 and Lemma 9 that the number of de Bruijn sequences constructed by the cycle joining method is equal to the number of spanning trees of its adjacency graph. Thus, by Lemma 8, the number of de Bruijn sequences constructed by the cycle joining method is $O(2^{3n})$ for odd n and $O(2^{7n})$ for even n. Note that the LFSR has $\frac{\phi(2^n-1)}{n}$ possible choices and $\frac{\phi(2^n-1)}{n}<\frac{2^n}{n}$, where ϕ is the Euler's phi-function [14]. Therefore, for odd n, the size of Ψ_{n+3} is $O(2^{3n}\frac{\phi(2^n-1)}{n})=O(2^{4n})$. Similarly, for even n, the size of Ψ_{n+3} is $O(2^{7n}\frac{\phi(2^n-1)}{n})=O(2^{8n})$. ■

By modifying the corresponding LFSRs, one can generate the de Bruijn sequences in Ψ_{n+3} through maximum-length NFSRs. To this end, some notation and results are introduced. Let $f(x_0,x_1,\ldots,x_{n-1})$ be the feedback function of an n-stage FSR and let

$$h(x_0,x_1,\ldots,x_{n-1})=f(x_0,x_1,\ldots,x_{n-1})+x_1^{a_1}x_2^{a_2}\ldots x_{n-1}^{a_{n-1}}, \tag{26}$$

where $(a_1,a_2,\ldots,a_{n-1})\in\mathbb{F}_2^{n-1}$, and $x_i^{a_i}=x_i+a_i+1$ for $1\le i\le n-1$. Note that $x_1^{a_1}x_2^{a_2}\ldots x_{n-1}^{a_{n-1}}=1$ if and only if $x_i=a_i$ for $1\le i\le n-1$. The values of the two functions h and f differ only at two conjugate states $\mathbf{A}=(a_0,a_1,\ldots,a_{n-1})$ and $\widehat{\mathbf{A}}=(\overline{a}_0,a_1,\ldots,a_{n-1})$. Then we have the following lemma.

Lemma 10 ([14]): Let $\Omega(f)$ and $\Omega(h)$ be the set of cycles generated by two FSRs with the feedback functions f and h in (26), respectively. Then $\Omega(f)$ and $\Omega(h)$ can be obtained from each other by interchanging the successors of $\mathbf{A}$ and $\widehat{\mathbf{A}}$.

Recall that every spanning tree of an adjacency graph corresponds to a de Bruijn sequence, since this represents a choice of conjugate pairs that repeatedly join two cycles into one ending with exactly one cycle in $\Omega(f)$ [16]. Let T be a spanning tree of the adjacency graph of an $(n+3)$-stage LFSR in $\mathcal{F}$, where n is an integer with $n>2$. Each of the edges in T can be labeled with an $(n+2)$-tuple $(a_1,a_2,\ldots,a_{n+2})$, where the two endpoints of the edge correspond to the two cycles sharing the conjugate pair $\{(a_0,a_1,a_2,\ldots,a_{n+2}),(\overline{a}_0,a_1,a_2,\ldots,a_{n+2})\}$. Denote by $E(T)$ the set of labels of all the edges in T. With the notation defined in this paragraph, Lemma 10 leads to the following result.

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Proposition 7: For a positive integer n with $n > 2$, let $f(x_0, x_1, \ldots, x_{n+2})$ be the feedback function of an $(n+3)$-stage LFSR in $\mathcal{F}$. Let $g(x_0, x_1, \ldots, x_{n+2}) = f(x_0, x_1, \ldots, x_{n+2}) + \sum_{(a_1,a_2,\ldots,a_{n+2})\in E(T)} x_1^{a_1} x_2^{a_2} \ldots x_{n+2}^{a_{n+2}}$, where T is a spanning tree of the adjacency graph of the LFSR. Then the NFSR with feedback function $g(x_0, x_1, \ldots, x_{n+2})$ generates de Bruijn sequences.

By Proposition 7, given a set $E(T)$, one can easily obtain an $(n+3)$-stage maximum-length NFSR. In the sequel we give an algorithm to generate a series of sets $E(T)$ for certain spanning trees.

For ease of presentation in Algorithm 1, the following expressions are used:

- Primitive Polynomial$(\mathbb{F}_2, n)$ creates a primitive polynomial of degree n in $\mathbb{F}_2[x]$;
- LFSR$(\mathbf{A}^*, p(x), N)$ outputs $(a_0, a_1, \ldots, a_{N-1})$ of the LFSR with characteristic polynomial $p(x)$ and initial state $\mathbf{A}^* = (a_0, a_1, \ldots, a_{n-1})$;
- Reduced(S) generates a set $\{(a_1, \ldots, a_{n+2}) \mid (a_0, a_1, \ldots, a_{n+2}) \in S\}$ for a subset S of $\mathbb{F}_2^{n+3}$.

Remark 3: Given a primitive polynomial $p(x)$ of degree n and an element $\mathbf{A}^* \in \mathbb{F}_2^n \setminus \{\mathbf{0}_n, \widehat{\mathbf{U}}_0^*\}$, the set $\mathcal{E}$ generated from Algorithm 1 has $|\mathcal{E}| = 8$ when n is odd, and $|\mathcal{E}| = 18$ when n is even. Since there are $2^n - 2$ distinct elements $\mathbf{A}^* \in \mathbb{F}_2^n \setminus \{\mathbf{0}_n, \widehat{\mathbf{U}}_0^*\}$, Algorithm 1 can be used to generate $8(2^n-2)$ distinct sets E when n is odd, and $18(2^n-2)$ distinct sets E when n is even.

In addition, it is worth pointing out that the time and memory complexity of Algorithm 1 are both $O(n)$ since each step in Algorithm 1 proceeds with bit-operations in constant/linear time with constant/linear bit-storage.

Theorem 4: Let n be an integer with $n > 2$. Then we have the following results.

(i) Given a primitive polynomial $p(x)$ of degree n and an element $\mathbf{A}^* \in \mathbb{F}_2^n \setminus \{\mathbf{0}_n, \widehat{\mathbf{U}}_0^*\}$, each E in $\mathcal{E}$ generated from Algorithm 1 is a set $E(T)$ for some spanning tree T of the adjacency graph of the LFSR with characteristic polynomial $(1+x^3)p(x)$.

(ii) The time and memory complexity of Algorithm 1 are both $O(n)$.

Proof: See Appendix B. ∎

Theorem 4 together with Proposition 7 immediately gives the following result.

Corollary 3: Let E be a set generated from Algorithm 1 for a primitive polynomial $p(x)$ of degree $n > 2$ and $f(x_0, x_1, \ldots, x_{n+2})$ be the feedback function of the LFSR with characteristic polynomial $(1+x^3)p(x)$. Then the NFSR with feedback function

$$f(x_0, x_1, \ldots, x_{n+2}) + \sum_{(a_1,a_2,\ldots,a_{n+2})\in E} x_1^{a_1} x_2^{a_2} \ldots x_{n+2}^{a_{n+2}} \quad (27)$$

generates de Bruijn sequences.

An example of the case $n = 3$ is given to illustrate that the sets E obtained from Algorithm 1 can be used to generate de Bruijn sequences.

Example 4: Consider the 6-stage LFSR in $\mathcal{F}$ obtained by setting the primitive polynomial $p(x) = x^3 + x^2 + 1 \in \mathbb{F}_2[x]$.

Algorithm 1 Generation of $\mathcal{E}$

Precomputation
$p(x) \leftarrow$ PrimitivePolynomial$(\mathbb{F}_2, n)$, $(u_0, u_1, u_2) \leftarrow (1, 0, 0)$
for $i \in \{0, 1, 2\}$ **do**
  $\mathcal{U} \leftarrow (u_i, u_{i+1}, \ldots, u_{i+n+2})$, where $u_{i+k} = u_{(i+k \bmod 3)}$ for $k \geq 3$
  $\widehat{\mathcal{U}} \leftarrow (\overline{u}_i, u_{i+1}, \ldots, u_{i+n+2})$,
  $\widehat{\mathcal{U}}^* \leftarrow (\overline{u}_i, u_{i+1}, \ldots, u_{i+n-1})$
  if $\widehat{\mathcal{U}} =$ LFSR$(\widehat{\mathcal{U}}^*, p(x), n+3)$ **then**
    $\mathbf{U}_0 \leftarrow (u_i, u_{i+1}, \ldots, u_{i+n+2})$
    $\widehat{\mathbf{U}}_0^* \leftarrow (\overline{u}_i, u_{i+1}, \ldots, u_{i+n-1})$
    $\mathbf{U}_1 \leftarrow (u_{i+1}, u_{i+2}, \ldots, u_{i+n+3})$
    $\mathbf{U}_2 \leftarrow (u_{i+2}, u_{i+3}, \ldots, u_{i+n+4})$
    break
  end if
end for
Main Algorithm
INPUT: an element $\mathbf{A}^* \in \mathbb{F}_2^n \setminus \{\mathbf{0}_n, \widehat{\mathbf{U}}_0^*\}$
  // $2^n - 2$ different choices
OUTPUT: a set $\mathcal{E}$ composed of different E

$\mathcal{E} \leftarrow \emptyset$, $E \leftarrow \emptyset$
$(a_0, a_1, \ldots, a_{n+4}) \leftarrow$ LFSR$(\mathbf{A}^*, p(x), n+5)$
if n is odd **then**
  $\mathbf{A} \leftarrow (a_0, a_1, \ldots, a_{n+2})$
  for $i, j, k \in \{1, 2\}$ **do**
    $E \leftarrow$ Reduced$\{\mathbf{0}_{n+3}, \mathbf{1}_{n+3}, \mathbf{A}, \overline{\mathbf{U}}_i, \mathbf{U}_j + \mathbf{A}, \mathbf{U}_k, \overline{\mathbf{A}}\}$
    $\mathcal{E} \leftarrow \mathcal{E} \cup \{E\}$
  end for
else if n is even **then**
  for $i \in \{0, 1, 2\}$ **do**
    $\mathbf{A}_i \leftarrow (a_i, a_{i+1}, \ldots, a_{i+n+2})$
  end for
  for $i, j \in \{0, 1, 2\}$ and $k \in \{1, 2\}$ **do**
    $E \leftarrow$ Reduced $\{\mathbf{0}_{n+3}, \mathbf{1}_{n+3}, \mathbf{U}_i, \overline{\mathbf{U}}_j, \mathbf{U}_k + \mathbf{A}_0, \mathbf{U}_0 + \mathbf{A}_0, \mathbf{U}_0 + \mathbf{A}_1, \mathbf{U}_0 + \mathbf{A}_2, \overline{\mathbf{U}}_0 + \mathbf{A}_0, \overline{\mathbf{U}}_0 + \mathbf{A}_1, \overline{\mathbf{U}}_0 + \mathbf{A}_2\}$
    $\mathcal{E} \leftarrow \mathcal{E} \cup \{E\}$
  end for
end if

return $\mathcal{E}$

Then the feedback function of the LFSR is $f(x_0, x_1, x_2, x_3, x_4, x_5) = x_0 + x_2 + x_5$. Taking $\mathbf{A}^* = (1, 1, 1)$, one obtains the following 8 sets E from Algorithm 1 (each 5-tuple $(a_1, a_2, \ldots, a_5) \in E$ is denoted by an integer $a = \sum_{i=1}^{5} a_i 2^{i-1}$ for ease of notation):

$\{0, 2, 4, 11, 20, 22, 31\}$, $\{0, 9, 11, 15, 20, 27, 31\}$,
$\{0, 2, 9, 11, 20, 22, 31\}$, $\{0, 2, 4, 11, 20, 27, 31\}$,
$\{0, 9, 11, 15, 20, 22, 31\}$, $\{0, 4, 11, 15, 20, 27, 31\}$,
$\{0, 2, 9, 11, 20, 27, 31\}$, $\{0, 4, 11, 15, 20, 22, 31\}$.

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With the above sets, we derive 8 feedback functions $g(x_0, x_1, \ldots, x_5)$ as in (27) and the following de Bruijn sequences with the initial state $(0, 0, 0, 0, 0, 0)$:

$$\begin{aligned}
&(00000011110111001101101001110101\\
&\quad 00011001011000101011111100001001),\\
&(00000011110000100011001001011000\\
&\quad 10101111110110111001101001110101),\\
&(00000011110111001101101001110101\\
&\quad 00011001001011000101011111100001),\\
&(00000011110110111001101001110101\\
&\quad 00011001011000101011111100001001),\\
&(00000011110000100011001001011000\\
&\quad 10101111110111001101101001110101),\\
&(00000011110000100100011001011000\\
&\quad 10101111110110111001101001110101),\\
&(00000011110110111001101001110101\\
&\quad 00011001001011000101011111100001),\\
&(00000011110000100100011001011000\\
&\quad 10101111110111001101101001110101).
\end{aligned}$$

Note that the first one of the eight sequences shown here is exactly the de Bruijn sequence given in (1), and the feedback function

$$\begin{aligned}
&g(x_0, x_1, x_2, x_3, x_4, x_5)\\
&\quad = x_0 + x_2 + x_5 + \overline{x}_1\overline{x}_2\overline{x}_3\overline{x}_4\overline{x}_5\\
&\qquad + \overline{x}_1x_2\overline{x}_3\overline{x}_4\overline{x}_5 + \overline{x}_1\overline{x}_2x_3\overline{x}_4\overline{x}_5\\
&\qquad + x_1x_2\overline{x}_3x_4\overline{x}_5 + \overline{x}_1\overline{x}_2x_3\overline{x}_4x_5\\
&\qquad + \overline{x}_1x_2x_3\overline{x}_4x_5 + x_1x_2x_3x_4x_5,
\end{aligned}$$

where $\overline{x}_i$ is the binary complement of x_i for $1 \le i \le 5$.

It should be pointed out that Algorithm 1 can be further generalized. Indeed, one can obtain a (reduced) simple graph of the adjacency graph of an LFSR in $\mathcal{F}$ by reducing the multiplicities of all multiple edges to one. Since the simple graph is static and has a small number of vertices, it is straightforward to precompute all its the spanning trees exhaustively. Let $\mathcal{T}$ be a certain spanning tree of the simple graph derived from Fig. 2 (resp. Fig. 3). Then, by Table I (resp. Table II) in Appendix B, each edge of $\mathcal{T}$ corresponds to a set of conjugate pairs with a common expression. Furthermore, for a fixed spanning tree $\mathcal{T}$ of the simple graph derived from Fig. 2 (or Fig. 3), one can generate a large class of spanning trees T of the corresponding adjacency graph, where each edge of a spanning tree T is obtained by explicitly taking one of the conjugate pairs sharing a common expression. In this way, one could design a generalized version of Algorithm 1, which can be used to generate all sets $E(T)$ corresponding to adjacency graphs in Figs. 2 and 3.

From Tables I and II, the most time-consuming steps in the generalized version of Algorithm 1 are to produce some states on the cycle $[\mathbf{s}]$ in $\Omega(q(x))$ satisfying specific conditions. When n is odd, since there is no restriction on the state $\mathbf{S}_j$, the steps that generate $\mathbf{S}_j$ can be proceeded with bit-operations in linear time with linear bit-storage. While in the case that n is even, some of the states are closely related to the permutation $Z(\cdot)$ over J^* and it seems that in these subcases an additional lookup table of $Z(\cdot)$ with memory complexity $O(2^n)$ is needed. Hence, it would be interesting to find more efficient ways to generate the states satisfying some conditions related to the permutation $Z(\cdot)$ over J^* for future research.

VI. Conclusion

In this paper, we determine the cycle structure and adjacency graphs of a class $\mathcal{F}$ of LFSRs. A new class of de Bruijn sequences is constructed from these LFSRs by the cycle joining method. For future research, it is possible to investigate some generalizations of the class $\mathcal{F}$.

Appendix A
Proofs of Several Results in Section IV

1) *Proof of Lemma 5:* Suppose that $\deg(f_1(x)) = m$ and $\mathrm{per}(\mathbf{a}) = l$, where both m and l are positive integers.

(i) To obtain a contradiction, we suppose that the cycle $[\mathbf{a}]$ in $\Omega(f(x))$ contains a conjugate pair $\{(a_i, a_{i+1}, \ldots, a_{i+n-1}), (a_j, a_{j+1}, \ldots, a_{j+n-1})\}$, where i, j are two integers with $0 \le i < j \le l-1$. By Equality (2), we have $(a_j, a_{j+1}, \ldots, a_{j+n-1}) = (\overline{a}_i, a_{i+1}, \ldots, a_{i+n-1})$. The condition that $f_1(x)$ is a proper factor of $f(x)$ implies that $m \le n-1$. Then

$$(a_{j+1}, a_{j+2}, \ldots, a_{j+m}) = (a_{i+1}, a_{i+2}, \ldots, a_{i+m}).$$

Thus, two sequences $L^{i+1}\mathbf{a}$ and $L^{j+1}\mathbf{a}$ in $\Omega(f_1(x))$ have the same initial state $(a_{i+1}, a_{i+2}, \ldots, a_{i+m})$. It follows that $L^{i+1}\mathbf{a} = L^{j+1}\mathbf{a}$. Then we have $\mathrm{per}(\mathbf{a}) \le j - i < l$, a contradiction. Thus, the first statement is proved.

(ii) The second statement can be proved in the similar way. Let k be a positive integer and $\mathrm{per}(\mathbf{b}) = k$. Suppose that $[\mathbf{a}]$ and $[\mathbf{b}]$ share a conjugate pair $\{(a_i, a_{i+1}, \ldots, a_{i+n-1}), (b_j, b_{j+1}, \ldots, b_{j+n-1})\}$, where i, j are two integers with $0 \le i \le l-1$ and $0 \le j \le k-1$. Then $(b_j, b_{j+1}, \ldots, b_{j+n-1}) = (\overline{a}_i, a_{i+1}, \ldots, a_{i+n-1})$. Similarly, we have $L^{i+1}\mathbf{a} = L^{j+1}\mathbf{b}$. This contradicts the condition that $[\mathbf{a}] \cap [\mathbf{b}] = \emptyset$. Thus, the second statement follows. ■

2) *Proof of Lemma 7:* Note that the m-sequence $\mathbf{s} = (s_0 s_1 \ldots s_{2^n-2})$ contains each nonzero binary n-tuple exactly once. Then $\widetilde{\mathbf{S}}_0 + \widetilde{\mathbf{S}}_j$ is a nonzero n-tuple due to $j \ne 0$ and it is a state on the cycle $[\mathbf{s}]$ different from the states $\widetilde{\mathbf{S}}_0$ and $\widetilde{\mathbf{S}}_j$. This leads to the first statement.

Suppose that $\mathbf{s}$ satisfies the linear recurrence $s_{i+n} = \sum_{t=0}^{n-1} c_t s_{i+t}$ for all $i \ge 0$. The equality $\widetilde{\mathbf{S}}_0 + \widetilde{\mathbf{S}}_j = \widetilde{\mathbf{S}}_k$ implies that $s_l = s_{j+l} + s_{k+l}$ for $l = 0, 1, \ldots, n-1$. Then

$$\begin{aligned}
s_n &= \sum_{t=0}^{n-1} c_t s_t = \sum_{t=0}^{n-1} c_t (s_{j+t} + s_{k+t})\\
&= \sum_{t=0}^{n-1} c_t s_{j+t} + \sum_{t=0}^{n-1} c_t s_{k+t} = s_{j+n} + s_{k+n}.
\end{aligned}$$

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That is, $s_l = s_{j+l} + s_{k+l}$ holds for $l = n$. The fact that $s_l = s_{j+l} + s_{k+l}$ holds for $l > n$ can be obtained in a similar way. It follows that $\mathbf{S}_0 + \mathbf{S}_j = \mathbf{S}_k$. ■

3) *Proof of Proposition 6:* It follows from Equalities (21) and (22) that $N(x, y) = |\widehat{\Phi}_x \cap \overline{\Phi}_y|$. By Equalities (21)–(23), we have $\widehat{\Phi}_x = \widehat{\Phi}_x^* \cup \{\widehat{\mathbf{U}}_x + \mathbf{S}_0\}$ and $\overline{\Phi}_y = \overline{\Phi}_y^* \cup \{\overline{\mathbf{U}}_y + \mathbf{S}_0\}$. Therefore, we have

$$\begin{aligned}\widehat{\Phi}_x \cap \overline{\Phi}_y &= \left(\widehat{\Phi}_x^* \cup \{\widehat{\mathbf{U}}_x + \mathbf{S}_0\}\right) \cap \left(\overline{\Phi}_y^* \cup \{\overline{\mathbf{U}}_y + \mathbf{S}_0\}\right) \\ &= \left(\widehat{\Phi}_x^* \cap \overline{\Phi}_y^*\right) \cup \left(\{\widehat{\mathbf{U}}_x + \mathbf{S}_0\} \cap \overline{\Phi}_y^*\right) \\ &\quad \cup \left(\widehat{\Phi}_x^* \cap \{\overline{\mathbf{U}}_y + \mathbf{S}_0\}\right). \end{aligned} \tag{28}$$

Denote by $\mathbf{0}_{n+3}$ the all-zero $(n+3)$-tuple. Then, by Equality (24), we have $\widehat{\mathbf{U}}_x + \mathbf{S}_0 \in \{\mathbf{0}_{n+3}, \overline{\mathbf{U}}_2, \overline{\mathbf{U}}_1\}$ and $\overline{\mathbf{U}}_y + \mathbf{S}_0 \in \{\overline{\mathbf{E}}, \widehat{\mathbf{U}}_2, \widehat{\mathbf{U}}_1\}$. Note that each state in $\{\mathbf{0}_{n+3}, \overline{\mathbf{U}}_2, \overline{\mathbf{U}}_1\}$ is on the cycle $[\mathbf{0}]$ or $[\overline{\mathbf{u}}]$. Since $\overline{\Phi}_y^*$ consists of states on the cycle $[L^y\overline{\mathbf{u}} + \mathbf{s}]$ and each of $[\mathbf{0}]$ and $[\overline{\mathbf{u}}]$ is disjoint with $[L^y\overline{\mathbf{u}}+\mathbf{s}]$, we have $\widehat{\mathbf{U}}_x+\mathbf{S}_0 \notin \overline{\Phi}_y^*$ for all $x, y \in \{0, 1, 2\}$. On the other hand, each state in $\{\overline{\mathbf{E}}, \widehat{\mathbf{U}}_2, \widehat{\mathbf{U}}_1\}$ is exactly the conjugate of some state on the cycle $[\mathbf{1}]$ or $[\mathbf{u}]$. For each $x \in \{0, 1, 2\}$, since any one state of $\widehat{\Phi}_x^*$ is the conjugate of some state on the cycle $[L^x\mathbf{u} + \mathbf{s}]$, we have $\overline{\mathbf{U}}_y + \mathbf{S}_0 \notin \widehat{\Phi}_x^*$ for all $y \in \{0, 1, 2\}$. Hence, by Equality (28), $N(x, y) = |\widehat{\Phi}_x \cap \overline{\Phi}_y| = |\widehat{\Phi}_x^* \cap \overline{\Phi}_y^*|$.

Note that $\Phi_i^* = \bigcup_{i'=0}^{2} \Phi_{i,i'}^*$ for each $i \in \{0, 1, 2\}$ with $\Phi_{i,i_1'}^* \cap \Phi_{i,i_2'}^* = \emptyset$ if $i_1' \neq i_2'$. Then

$$\begin{aligned} N(x, y) &= \Big|(\bigcup_{i'=0}^{2} \widehat{\Phi}_{x,i'}^*) \cap \overline{\Phi}_y^*\Big| = \Big|\bigcup_{i'=0}^{2} (\widehat{\Phi}_{x,i'}^* \cap \overline{\Phi}_y^*)\Big| \\ &= \sum_{i'=0}^{2} \left|\widehat{\Phi}_{x,i'}^* \cap \overline{\Phi}_y^*\right| = \left|\widehat{\Phi}_{x,3-x}^* \cap \overline{\Phi}_y^*\right| \\ &\quad + \left|\widehat{\Phi}_{x,2-x}^* \cap \overline{\Phi}_y^*\right| + \left|\widehat{\Phi}_{x,1-x}^* \cap \overline{\Phi}_y^*\right|, \end{aligned} \tag{29}$$

where the subscript $i - x$ is modulo 3 for each $i \in \{0, 1, 2\}$. To count the sizes of the intersections, the three sets $\Phi_{x,3-x}^*$, $\Phi_{x,2-x}^*$ and $\Phi_{x,1-x}^*$ are characterized separately.

By Equalities (18) and (25),

$$\begin{aligned}\Phi_{x,3-x}^* &= \left\{\mathbf{U}_0 + \mathbf{S}_j \mid j \equiv (3-x) \pmod 3,\ j \in J^*\right\} \\ &= \left\{\widehat{\mathbf{S}}_{Z(j)} \mid j \equiv (3-x) \pmod 3,\ j \in J^*\right\}. \end{aligned} \tag{30}$$

Then $\widehat{\Phi}_{x,3-x}^* = \left\{\mathbf{S}_{Z(j)} \mid j \equiv (3-x) \pmod 3,\ j \in J^*\right\}$. That is, $\widehat{\Phi}_{x,3-x}^*$ consists of states on the cycle $[\mathbf{s}]$. Since $\overline{\Phi}_y^*$ consists of states on the cycle $[L^y\overline{\mathbf{u}}+\mathbf{s}]$, and two cycles $[\mathbf{s}]$, $[L^y\overline{\mathbf{u}}+\mathbf{s}]$ are disjoint, we have $\left|\widehat{\Phi}_{x,3-x}^* \cap \overline{\Phi}_y^*\right| = 0$.

Note that $\mathbf{U}_l + \mathbf{U}_0 = \overline{\mathbf{U}}_{3-l}$ holds for $l \in \{1, 2\}$. Similarly to Equality (30), we have

$$\begin{aligned}\Phi_{x,2-x}^* &= \left\{\mathbf{U}_2 + \mathbf{S}_j \mid j \equiv (2-x) \pmod 3,\ j \in J^*\right\} \\ &= \left\{\overline{\mathbf{U}}_1 + \mathbf{U}_0 + \mathbf{S}_j \mid j \equiv (2-x) \pmod 3,\ j \in J^*\right\} \\ &= \left\{\overline{\mathbf{U}}_1 + \widehat{\mathbf{S}}_{Z(j)} \mid j \equiv (2-x) \pmod 3,\ j \in J^*\right\}. \end{aligned}$$

Then $\widehat{\Phi}_{x,2-x}^* = \left\{\overline{\mathbf{U}}_1 + \mathbf{S}_{Z(j)} \mid j \equiv (2-x) \pmod 3,\ j \in J^*\right\}$. Note that $\overline{\Phi}_y^* = \left\{\overline{\mathbf{U}}_{y+j} + \mathbf{S}_j \mid j \in J^*\right\}$.

Let $\mathbf{A} = \overline{\mathbf{U}}_{y+j_1} + \mathbf{S}_{j_1}$ for a fixed $j_1 \in J^*$. If there exists an integer $l \in J^*$ such that $l \equiv (2-x) \pmod 3$ and $y + j_1 = y + Z(l) \equiv 1 \pmod 3$, then $\mathbf{A} \in \widehat{\Phi}_{x,2-x}^* \cap \overline{\Phi}_y^*$. Conversely, if $\mathbf{A} \in \widehat{\Phi}_{x,2-x}^* \cap \overline{\Phi}_y^*$, then $\mathbf{A} = \overline{\mathbf{U}}_{y+j_1} + \mathbf{S}_{j_1} = \overline{\mathbf{U}}_1 + \mathbf{S}_{Z(j_2)}$ for some integer $j_2 \in J^*$ with $j_2 \equiv (2-x) \pmod 3$. It follows that $\overline{\mathbf{U}}_{y+j_1} + \mathbf{S}_{j_1} = \overline{\mathbf{U}}_1 + \mathbf{S}_{Z(j_2)}$, or equivalently,

$$\overline{\mathbf{U}}_{y+j_1} + \overline{\mathbf{U}}_1 = \mathbf{S}_{j_1} + \mathbf{S}_{Z(j_2)}. \tag{31}$$

Since $\mathbf{U}_l + \mathbf{U}_0 = \overline{\mathbf{U}}_{3-l}$ holds for $l \in \{1, 2\}$, we have $\overline{\mathbf{U}}_{y+j_1} + \overline{\mathbf{U}}_1 \in \{\mathbf{U}_2, \mathbf{0}_{n+3}, \overline{\mathbf{U}}_0\}$. If $j_1 \neq Z(j_2)$, it can be proved similarly as in Lemma 7 that there exists some $j' \in J^*$ such that $\mathbf{S}_{j_1} + \mathbf{S}_{Z(j_2)} = \mathbf{S}_{j'}$. From Equality (31), we have $j_1 = Z(j_2)$ and $\overline{\mathbf{U}}_{y+j_1} = \overline{\mathbf{U}}_1$. Thus, if $\mathbf{A} \in \widehat{\Phi}_{x,2-x}^* \cap \overline{\Phi}_y^*$, then there exists an integer $j_2 \in J^*$ such that $j_2 \equiv (2-x) \pmod 3$ and $y+Z(j_2) \equiv 1 \pmod 3$. Therefore, $\mathbf{A} \in \widehat{\Phi}_{x,2-x}^* \cap \overline{\Phi}_y^*$ if and only if there exists an integer $l \in J^*$ such that $l \equiv (2-x) \pmod 3$ and $y + j_1 = y + Z(l) \equiv 1 \pmod 3$. Consequently, $|\widehat{\Phi}_{x,2-x}^* \cap \overline{\Phi}_y^*|$ equals the number of pairs $(l, Z(l))$ such that $l \in J^*$, $l \equiv (2-x) \pmod 3$ and $y+Z(l) \equiv 1 \pmod 3$. Thus, by Lemma 1, $|\widehat{\Phi}_{x,2-x}^* \cap \overline{\Phi}_y^*| = (2-x, 1-y)_3$.

The equality $|\widehat{\Phi}_{x,1-x}^* \cap \overline{\Phi}_y^*| = (1-x, 2-y)_3$ can be obtained in the same manner. Hence by Equality (29), $N(x, y) = (2-x, 1-y)_3 + (1-x, 2-y)_3$. ■

4) *Proof of Corollary 2:* By Equality (30), we have $\widehat{\Phi}_{x,3-x}^* = \{\mathbf{S}_{Z(j)} \mid j \equiv (3-x) \pmod 3,\ j \in J^*\}$ for each $x \in \{0, 1, 2\}$. Then

$$\widehat{\Phi}_{0,0}^* \cup \widehat{\Phi}_{1,2}^* \cup \widehat{\Phi}_{2,1}^* = \left\{\mathbf{S}_{Z(j)} \mid j \in J^*\right\} = \left\{\mathbf{S}_j \mid j \in J^*\right\} \tag{32}$$

holds due to the fact that the mapping Z is a permutation of the set J^*. Since the size of $\widehat{\Phi}_{x,3-x}^*$ only depends on x, we have $|\widehat{\Phi}_{0,0}^*| = \frac{2^n-1}{3} - 1$ and $|\widehat{\Phi}_{1,2}^*| = |\widehat{\Phi}_{2,1}^*| = \frac{2^n-1}{3}$.

Note that $\widehat{\Phi}_{x,3-x}^*$ is the set of conjugates of states on the cycle $[L^x\mathbf{u} + \mathbf{s}]$ for each $x \in \{0, 1, 2\}$. It follows from Equalities (30) and (32) that the cycle $[\mathbf{s}]$ shares exactly $\frac{2^n-1}{3} - 1$ conjugate pairs with the cycle $[\mathbf{u} + \mathbf{s}]$ and $\frac{2^n-1}{3}$ conjugate pairs with each of the cycles $[L\mathbf{u}+\mathbf{s}]$ and $[L^2\mathbf{u}+\mathbf{s}]$. Thus, by Lemma 6, the cycle $[\overline{\mathbf{s}}]$ shares exactly $\frac{2^n-1}{3} - 1$ conjugate pairs with the cycle $[\overline{\mathbf{u}} + \mathbf{s}]$ and $\frac{2^n-1}{3}$ conjugate pairs with each of the cycles $[L\overline{\mathbf{u}} + \mathbf{s}]$ and $[L^2\overline{\mathbf{u}} + \mathbf{s}]$. ■

APPENDIX B
PROOF OF THEOREM 4

Proof: (i) With the same notation in Section III, we take $q(x) = (1+x^3)p(x)$ and denote by $\Omega(q(x))$ the set of all 2^{n+3} sequences generated by $q(x)$. Recall that the adjacency graph of the LFSR with characteristic polynomial $q(x)$ is given in Theorem 1 for odd n and in Theorem 2 for even n, respectively.

The equality $\mathbf{U}_0 + \mathbf{S_0} = \mathbf{E}$ in (12) implies that $\widehat{\mathbf{U}}_0 = \mathbf{S_0}$ is a state on the cycle $[\mathbf{s}]$. Thus one can fix the state $\mathbf{U}_0$ after confirming that its conjugate satisfies the linear recurrence of $p(x)$, and then determine the states $\mathbf{U}_1$ and $\mathbf{U}_2$ in Precomputation of Algorithm 1.

Case 1: In the case of odd n, by Propositions 3, 4 and Theorem 1, the conjugate pairs between adjacent cycles in $\Omega(q(x))$ are listed in Table I.

For the primitive polynomial $p(x)$ and the element $\mathbf{A}^* \in \mathbb{F}_2^n \setminus \{\mathbf{0}_n, \widehat{\mathbf{U}}_0^*\}$, Algorithm 1 generates a set $\mathcal{E}$ consisting

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TABLE I

THE ADJACENT CYCLES AND CONJUGATE PAIRS FOR ODD n

Cycles	Representatives of Conjugate pairs[1]
$[\mathbf{s}]$, $[\mathbf{u}]$	$\{\mathbf{U}_0\}$
$[\mathbf{0}]$, $[\mathbf{u}+\mathbf{s}]$	$\{\mathbf{0}_{n+3}\}$
$[\mathbf{s}]$, $[\mathbf{u}+\mathbf{s}]$	$\{\mathbf{S}_j \mid j \in J^*\}$
$[\overline{\mathbf{u}}]$, $[\mathbf{u}+\mathbf{s}]$	$\{\overline{\mathbf{U}}_1, \overline{\mathbf{U}}_2\}$
$[\overline{\mathbf{u}}+\mathbf{s}]$, $[\mathbf{u}+\mathbf{s}]$	$\{\mathbf{U}_i + \mathbf{S}_j \mid i \in \{1,2\}, j \in J^*\}$
$[\mathbf{u}]$, $[\overline{\mathbf{u}}+\mathbf{s}]$	$\{\mathbf{U}_1, \mathbf{U}_2\}$
$[\overline{\mathbf{s}}]$, $[\overline{\mathbf{u}}+\mathbf{s}]$	$\{\overline{\mathbf{S}}_j \mid j \in J^*\}$
$[\mathbf{1}]$, $[\overline{\mathbf{u}}+\mathbf{s}]$	$\{\mathbf{1}_{n+3}\}$
$[\overline{\mathbf{s}}]$, $[\overline{\mathbf{u}}]$	$\{\overline{\mathbf{U}}_0\}$

[1] In order to avoid cumbersomeness of notation, we present only one state (representative) of a conjugate pair in Table I since the other one can be easily obtained.

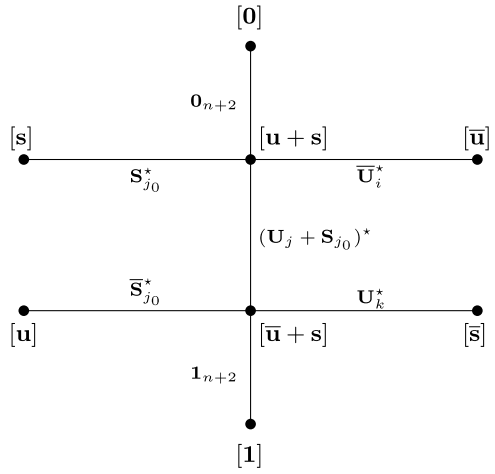


Fig. 4. A labeled spanning tree of the adjacency graph.

of several sets

$$E = \text{Reduced}\left\{\mathbf{0}_{n+3}, \mathbf{1}_{n+3}, \mathbf{A}, \overline{\mathbf{U}}_i, \mathbf{U}_j + \mathbf{A}, \mathbf{U}_k, \overline{\mathbf{A}}\right\}$$

for $i, j, k \in \{1, 2\}$, where $\mathbf{A} = \text{LFSR}(\mathbf{A}^*, p(x), n+3)$ and the expressions LFSR$(\cdot)$, Reduced$(\cdot)$ are introduced before Algorithm 1. It is readily seen that $\mathbf{A} = \text{LFSR}(\mathbf{A}^*, p(x), n+3)$ is a state on the cycle $[\mathbf{s}]$ in $\Omega(q(x))$ since $\mathbf{A}^* \neq \mathbf{0}_n$. Moreover, the fact $\mathbf{A}^* \neq \widehat{\mathbf{U}}_0^*$ implies $\mathbf{A} \neq \widehat{\mathbf{U}}_0$, i.e., $\mathbf{A} \neq \mathbf{S}_0$. Thus, there exists a certain $j_0 \in J^*$ such that $\mathbf{A} = \mathbf{S}_{j_0}$.

Denote $\mathbf{X}^\star = (x_1, \ldots, x_{n+2})$ for any $(n+3)$-tuple $\mathbf{X} = (x_0, x_1, \ldots, x_{n+2})$. By Table I, each element in E corresponds to a representative of a conjugate pair between two adjacent cycles in $\Omega(q(x))$. Therefore, for $i, j, k \in \{1, 2\}$, the set

$$E = \left\{\mathbf{0}_{n+2}, \mathbf{1}_{n+2}, \mathbf{S}_{j_0}^\star, \overline{\mathbf{U}}_i^\star, (\mathbf{U}_j + \mathbf{S}_{j_0})^\star, \mathbf{U}_k^\star, \overline{\mathbf{S}}_{j_0}^\star\right\} = E(T)$$

corresponds to the spanning tree T as shown in Fig. 4.

Case 2: In the case of even n, by Propositions 3, 5 and Theorem 2, the conjugate pairs between adjacent cycles in $\Omega(q(x))$ are listed in Table II.

For the primitive polynomial $p(x)$ and the element $\mathbf{A}^* \in \mathbb{F}_2^n \setminus \{\mathbf{0}_n, \widehat{\mathbf{U}}_0^*\}$, Algorithm 1 generates a set $\mathcal{E}$ consisting of

$$E = \text{Reduced}\left\{\mathbf{0}_{n+3}, \mathbf{1}_{n+3}, \mathbf{U}_i, \overline{\mathbf{U}}_j, \mathbf{U}_k + \mathbf{A}_0\right\} \cup \bigcup_{0 \le t \le 2} \text{Reduced}\{\mathbf{U}_0 + \mathbf{A}_t, \overline{\mathbf{U}}_0 + \mathbf{A}_t\}$$

for $i, j \in \{0, 1, 2\}$ and $k \in \{1, 2\}$.

TABLE II

THE ADJACENT CYCLES AND CONJUGATE PAIRS FOR EVEN n

Cycles	Conjugate pairs representatives[1]
$[\mathbf{0}]$, $[\mathbf{u}+\mathbf{s}]$	$\{\mathbf{0}_{n+3}\}$
$[\mathbf{s}]$, $[L^x\mathbf{u}+\mathbf{s}]^2$	$\{\mathbf{U}_0 + \mathbf{S}_j \mid j \equiv (-x) \pmod 3, j \in J^*\}$
$[\mathbf{s}]$, $[\mathbf{u}]$	$\{\mathbf{U}_0\}$
$[\mathbf{u}]$, $[L\overline{\mathbf{u}}+\mathbf{s}]$	$\{\mathbf{U}_2\}$
$[\mathbf{u}]$, $[L^2\overline{\mathbf{u}}+\mathbf{s}]$	$\{\mathbf{U}_1\}$
$[L^x\mathbf{u}+\mathbf{s}]$, $[L^y\overline{\mathbf{u}}+\mathbf{s}]^2$	$\{\mathbf{U}_i + \mathbf{S}_j \mid i \in \{1,2\}, j \in \Gamma(i,x,y)\}$ [3]
$[\overline{\mathbf{u}}]$, $[L^2\mathbf{u}+\mathbf{s}]$	$\{\overline{\mathbf{U}}_1\}$
$[\overline{\mathbf{u}}]$, $[L\mathbf{u}+\mathbf{s}]$	$\{\overline{\mathbf{U}}_2\}$
$[\overline{\mathbf{s}}]$, $[\overline{\mathbf{u}}]$	$\{\overline{\mathbf{U}}_0\}$
$[\overline{\mathbf{s}}]$, $[L^y\overline{\mathbf{u}}+\mathbf{s}]^2$	$\{\overline{\mathbf{U}}_0 + \mathbf{S}_j \mid j \equiv (-x) \pmod 3, j \in J^*\}$
$[\mathbf{1}]$, $[\overline{\mathbf{u}}+\mathbf{s}]$	$\{\mathbf{1}_{n+3}\}$

[1] Similarly to Table I, only one state (representative) of a conjugate pair is presented in Table II.

[2] $x, y \in \{0, 1, 2\}$

[3] $\Gamma(i,x,y) = \{j \in J^* \mid j \equiv (i-x) \pmod 3, Z(j) \equiv (-i-y) \pmod 3\}$.

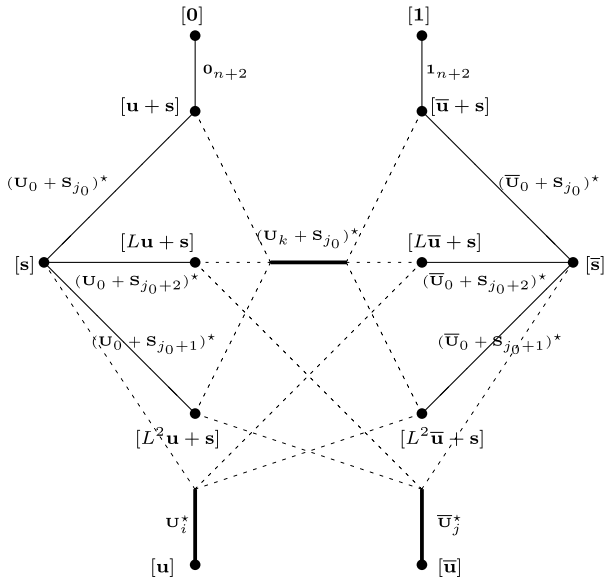


Fig. 5. A labeled spanning tree of the adjacency graph.

It follows from Table II that $\mathbf{0}_{n+3}$ is the representative of the conjugate pair between $[\mathbf{0}]$ and $[\mathbf{u}+\mathbf{s}]$, and $\mathbf{1}_{n+3}$ is the representative of the conjugate pair between $[\mathbf{1}]$ and $[\overline{\mathbf{u}}+\mathbf{s}]$. Similarly to the analysis for Case 1, one can obtain a certain $j_0 \in J^*$ such that $\mathbf{A}_0 = \mathbf{S}_{j_0}$.

We claim that each set E in $\mathcal{E}$ is a set $E(T)$ for the spanning tree T of the adjacency graph given in Fig. 5. The claim will be proved in three subcases.

Subcase 2.1: For $t = 0, 1, 2$, we have $\mathbf{A}_t = \mathbf{S}_{j_0+t}$. Thus, $\mathbf{A}_0$, $\mathbf{A}_1$ and $\mathbf{A}_2$ are three consecutive states on the cycle $[\mathbf{s}]$. Then, by Table II, the set $\bigcup_{0 \le t \le 2} \{\mathbf{U}_0 + \mathbf{S}_{j_0+t}\}$ exactly represents three conjugate pairs between the cycle $[\mathbf{s}]$ and three cycles $[\mathbf{u}+\mathbf{s}]$, $[L\mathbf{u}+\mathbf{s}]$ and $[L^2\mathbf{u}+\mathbf{s}]$. Likewise, the set $\bigcup_{0 \le t \le 2} \{\overline{\mathbf{U}}_0 + \mathbf{S}_{j_0+t}\}$ exactly represents three conjugate pairs between the cycle $[\overline{\mathbf{s}}]$ and three cycles $[\overline{\mathbf{u}}+\mathbf{s}]$, $[L\overline{\mathbf{u}}+\mathbf{s}]$ and $[L^2\overline{\mathbf{u}}+\mathbf{s}]$.

Subcase 2.2: Consider fixed $i, j \in \{0, 1, 2\}$. It follows from Table II that $\mathbf{U}_i$ is a representative of the conjugate pair shared by the cycle $[\mathbf{u}]$ and one of the three cycles $[\mathbf{s}]$, $[L\overline{\mathbf{u}}+\mathbf{s}]$ and

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$[L^2\overline{\mathbf{u}}+\mathbf{s}]$. Similarly, $\overline{\mathbf{U}}_j$ is a representative of the conjugate pair shared by the cycle $[\overline{\mathbf{u}}]$ and one of the three cycles $[\overline{\mathbf{s}}]$, $[L\mathbf{u}+\mathbf{s}]$ and $[L^2\mathbf{u}+\mathbf{s}]$.

Subcase 2.3: By Table II, the remaining element $\mathbf{U}_k+\mathbf{A}_0=\mathbf{U}_k+\mathbf{S}_{j_0}$ is a representative of a conjugate pair between two cycles $[L^x\mathbf{u}+\mathbf{s}]$ and $[L^y\overline{\mathbf{u}}+\mathbf{s}]$ if and only if

$$j_0 \equiv (k-x) \pmod 3,\ Z(j_0) \equiv (-k-y) \pmod 3.$$

Therefore, for fixed k, $\mathbf{U}_k+\mathbf{S}_{j_0}$ is a representative of a conjugate pair between two cycles $[L^x\mathbf{u}+\mathbf{s}]$ and $[L^y\overline{\mathbf{u}}+\mathbf{s}]$, where $x \equiv (k-j_0) \pmod 3$, $y \equiv (-k-Z(j_0)) \pmod 3$.

On combing the three subcases, it follows that for $i, j \in \{0,1,2\}$, $k \in \{1,2\}$,

$$E = \left\{\mathbf{0}_{n+2}, \mathbf{1}_{n+2}, \mathbf{U}_i^\star, \overline{\mathbf{U}}_j^\star, (\mathbf{U}_k+\mathbf{S}_{j_0})^\star\right\} \cup \bigcup_{0\le t\le 2} \{(\mathbf{U}_0+\mathbf{S}_{j_0+t})^\star, (\overline{\mathbf{U}}_0+\mathbf{S}_{j_0+t})^\star\},$$

and it represents the edges shown in Fig. 5. It is easy to see that Fig. 5 gives a spanning tree of the corresponding adjacency graph (see further explanations of the dash lines in Remark 4). The claim thus follows.

(ii) As pointed out in Remark 3, for each step in Algorithm 1, one only needs $O(n)$ bit-storage to proceed with $O(n)$ bit-operations. This implies that Algorithm 1 has time and memory complexity $O(n)$. ■

Remark 4: It should be pointed out that the dash lines and the regular line connecting the cycles $[\mathbf{u}]$, $[\mathbf{s}]$, $[L\overline{\mathbf{u}}+\mathbf{s}]$ and $[L^2\overline{\mathbf{u}}+\mathbf{s}]$ in Fig. 5 indicate there is an edge between the cycle $[\mathbf{u}]$ and one of the others three cycles in the spanning tree. Likewise, as indicated in Fig. 5, there exists an edge between the cycle $[\overline{\mathbf{u}}]$ and one cycle of $[\overline{\mathbf{s}}]$, $[L\mathbf{u}+\mathbf{s}]$ and $[L^2\mathbf{u}+\mathbf{s}]$. In addition, the dash lines and the regular line connecting the cycles $[L^x\mathbf{u}+\mathbf{s}]$ and $[L^y\overline{\mathbf{u}}+\mathbf{s}]$ for all $x,\ y \in [0,1,2]$ in Fig. 5 indicate the existence of an edge between two cycles of them for certain specified x, y.

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